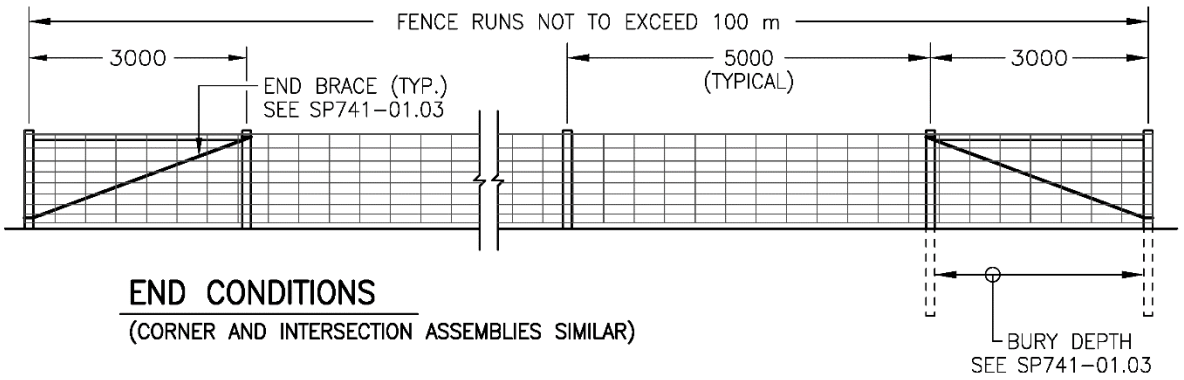
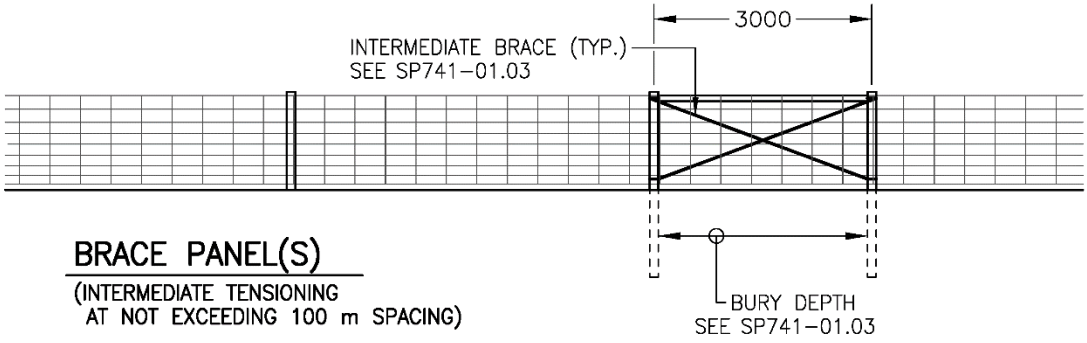
TYPE A, B & B1 FENCE STABLIIZATION REQUIREMENTS

SP741–01.02



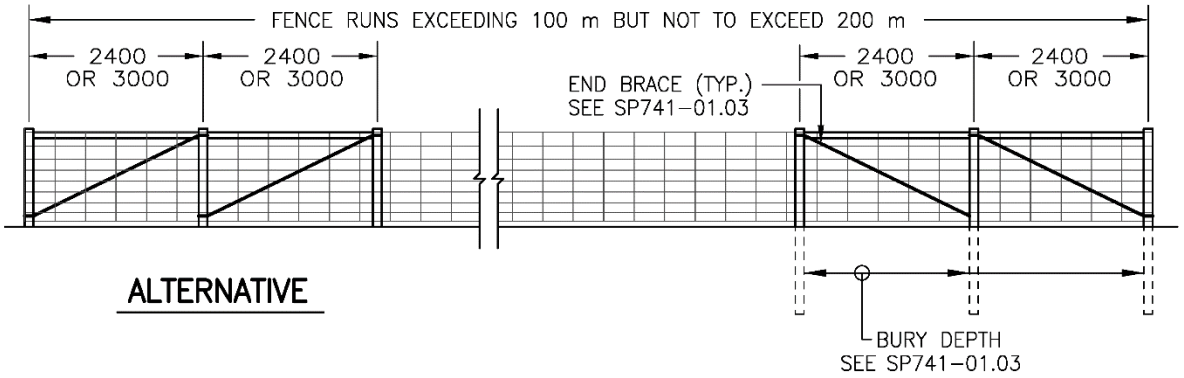
END CONDITIONS

(CORNER AND INTERSECTION ASSEMBLIES SIMILAR)

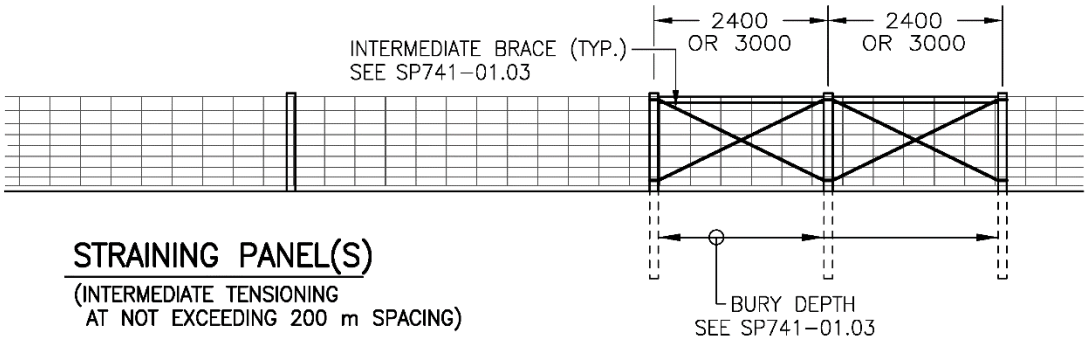


BRACE PANEL(S)

(INTERMEDIATE TENSIONING
AT NOT EXCEEDING 100 m SPACING)



ALTERNATIVE



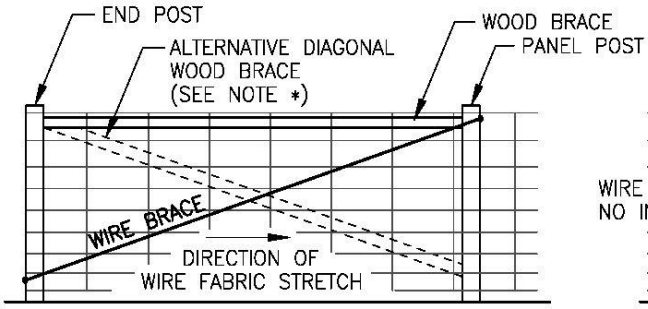
STRAINING PANEL(S)

(INTERMEDIATE TENSIONING
AT NOT EXCEEDING 200 m SPACING)

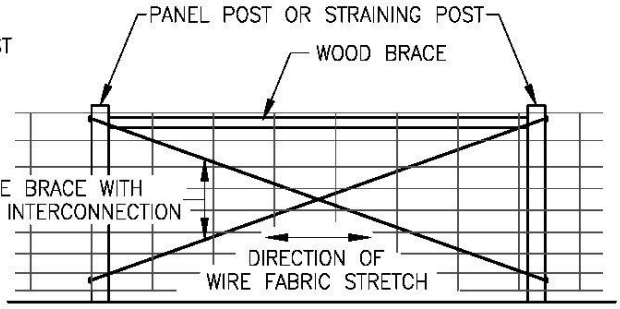
NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

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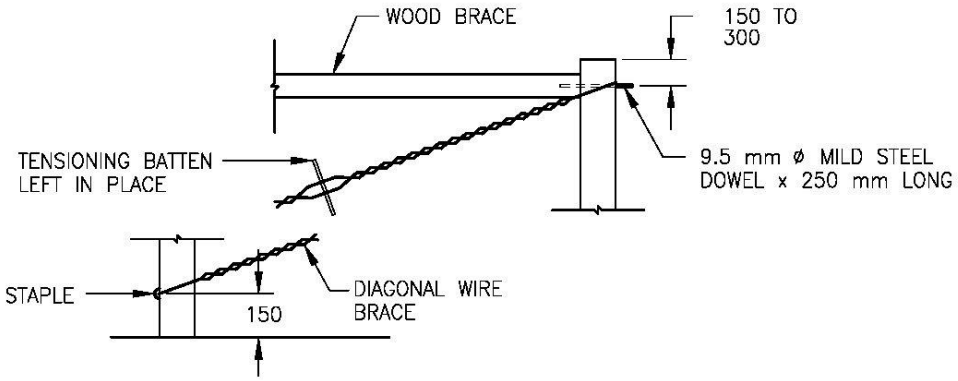
TYPE A, B & B1 FENCE STABILIZATION REQUIREMENTS | SP741–01.03



END BRACE
(BASIC END TENSIONING PANEL)



INTERMEDIATE BRACE
(BASIC INTERMEDIATE TENSIONING PANEL)



WOOD AND WIRE BRACE DETAIL

GENERAL NOTES:

WIRE FABRIC ROLL LENGTH 100 m.

STRAINING PANELS AND/OR BRACE PANELS ARE REQUIRED FOR TENSIONING FABRIC AND AT SIGNIFICANT CHANGES IN FENCE ALIGNMENT.

POST SIZES: ALL POSTS – 120 mm (MIN.) DIAMETER EXCEPT:
LINE POSTS – 100 mm (MIN.) DIAMETER
CORNER POSTS – 140 mm (MIN.) DIAMETER

POST LENGTH: (EXCEPT LINE POSTS AND GATE POSTS) ALL POSTS TO BE THE SPECIFIED LINE POST LENGTH (SEE SP741–01.01) PLUS 400mm FOR THE STABILIZING INCREASED DEPTH OF BURY.

BURY DEPTH: DEPTH OF END AND PANEL POSTS NORMALLY NOT LESS THAN 1200 mm. WHERE 1200 mm NOT ATTAINABLE BUT EXCEEDING 900 mm, USE DOUBLE TENSIONING STRAINING PANEL ASSEMBLIES AT NOT EXCEEDING 200 m SPACING. GREATER MINIMUMS ARE REQUIRED FOR PEAT SOILS, ETC.

POST SPACING AND WOOD BRACES:

WOOD BRACE	SIZE	LENGTH	POST SPACING APPROXIMATE
HORIZONTAL	80 mm	2.4 m	2.5 m
HORIZONTAL	100 mm	3.0 m	3.1 m
* DIAGONAL	100 mm	3.0 m	2.9 m

* NOTE: SOFT, MARSH AND/OR PEAT SOIL CONDITIONS MAY WARRANT DIAGONAL WOOD BRACES IN PLACE OF HORIZONTALS IN addition TO THE LONGER POSTS.

NOTE: ROCK, SHALE OR FRIABLE ROCK CONDITIONS SEE SP741–04.01 FOR METAL POSTS AND BRACES.

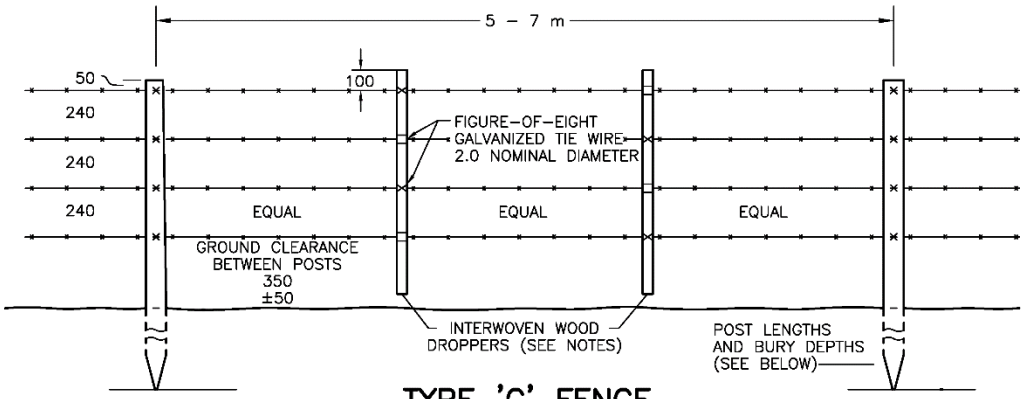
WIRE BRACES: 2 STRANDS OF 3.75 mm NOMINAL DIAMETER GALVANIZED WIRE WITH LOOPED ENDS AROUND DOWEL AND STAPLED TO POSTS – TWISTED TAUT ONE WAY OR

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

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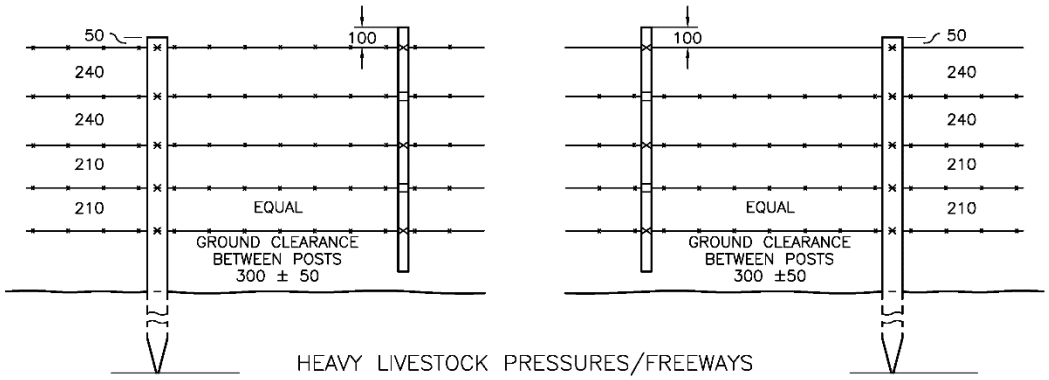
TYPE C, C1, & C2 STANDARD WIRE FENCE

SP741–02.01



TYPE 'C' FENCE

(BARBED WIRE)



TYPE 'C1' FENCE

(BARBED WIRE)

TYPE 'C2' FENCE

(WILD–LIFE SENSITIVE AREAS)

NOTES:

1. FENCE CONSTRUCTION SHALL BE IN ACCORDANCE WITH SECTION 741 OR AS OTHERWISE SPECIFIED OR REQUIRED.
2. PRESSURE TREATED WOOD POST AND BRACE MATERIAL AS SPECIFIED IN SECTION 909.
3. DOUBLE STRANDED GALVANIZED BARBED WIRE AS SPECIFIED BY ASTM A 121 AND SUBSECTION 316.06 FOR TYPE 'C' & 'C1' FENCE.
4. SINGLE STRAND HIGH–TENSILE WIRE AS SPECIFIED IN SUBSECTION 316.07 FOR TYPE 'C2' FENCE.
5. STAPLES: 45 mm LONG OF 3.5 mm NOMINAL DIAMETER GALVANIZED WIRE TO EVERY LINE OF BARBED AND SMOOTH WIRE.
6. DROPPERS: WOOD 50 mm DIAMETER OR OTHER ALTERNATIVE SPECIFIED BY SUBSECTION 909.08.
7. LINE POSTS: 100 mm MINIMUM DIAMETER POINTED FOR DRIVING SET PLUMB AND FIRM.

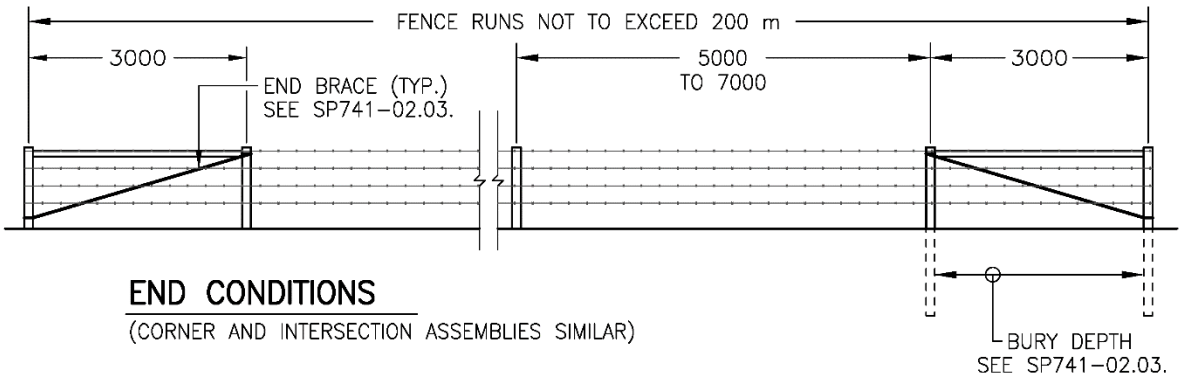
BURY DEPTH (mm)	POST LENGTH (m)	
	TYPE 'C'	TYPE 'C1 & C2'
STANDARD 850	2.0	2.1
PEAT SOIL 1050	2.2	2.3

8. ROCK, SHALE AND FRIABLE ROCK CONDITIONS WARRANT METAL POSTS – SEE SP741–04.01 FOR MINOR ROCK OUTCROPPINGS AND UNFORESEEN MINIMUM OVERBURDEN OCCURENCES, WIRE TIED MULTIPLE WOOD POST/BRACE ASSEMBLIES MAY BE PERMITTED BY THE MINISTRY REPRESENTATIVE.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

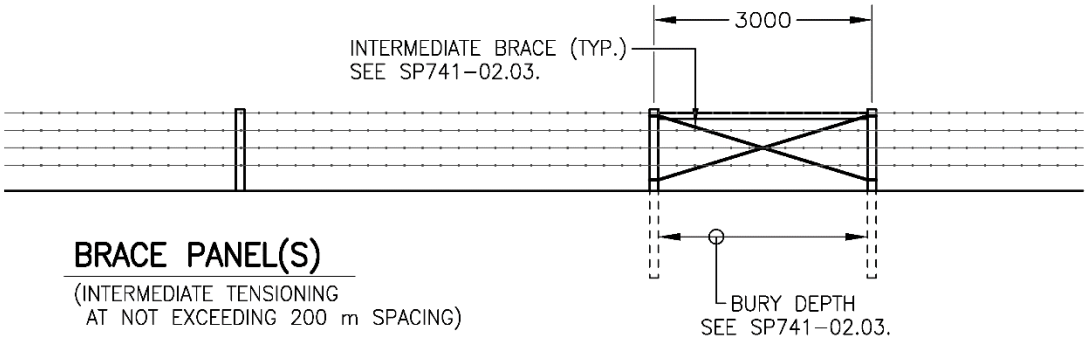
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TYPE C, C1 & C2 FENCE STABILIZATION REQUIREMENTS | SP741–02.02



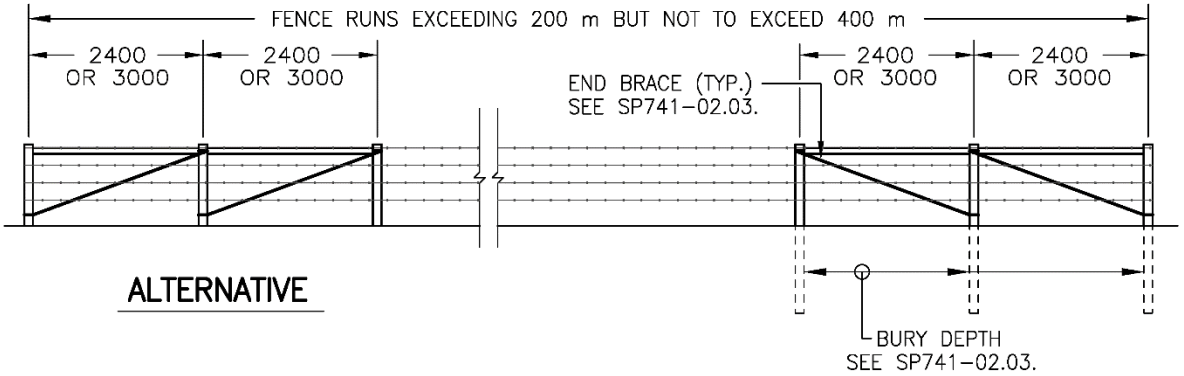
END CONDITIONS

(CORNER AND INTERSECTION ASSEMBLIES SIMILAR)

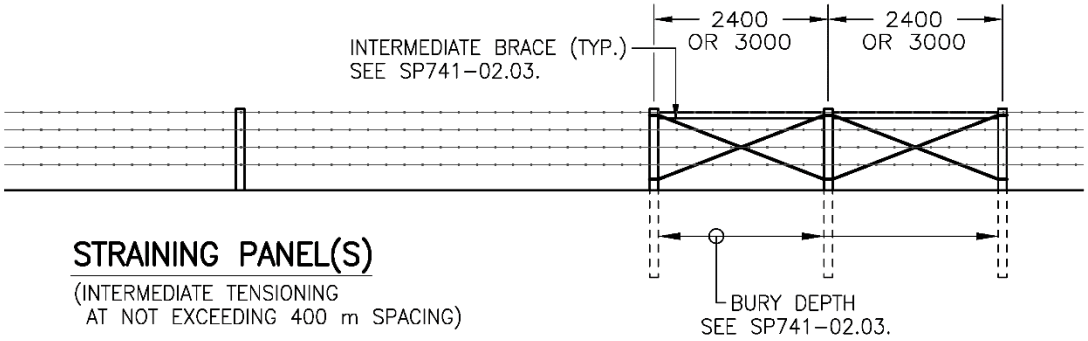


BRACE PANEL(S)

(INTERMEDIATE TENSIONING
AT NOT EXCEEDING 200 m SPACING)



ALTERNATIVE



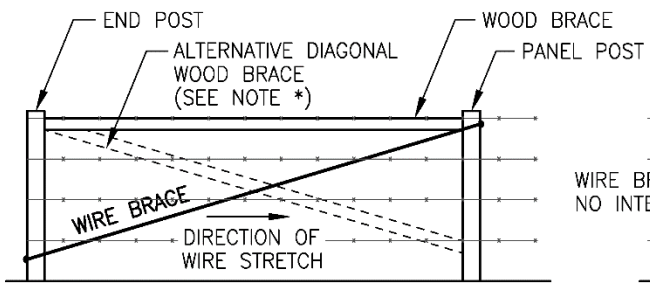
STRAINING PANEL(S)

(INTERMEDIATE TENSIONING
AT NOT EXCEEDING 400 m SPACING)

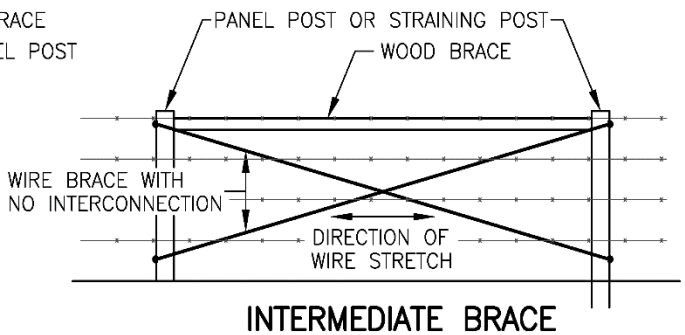
NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

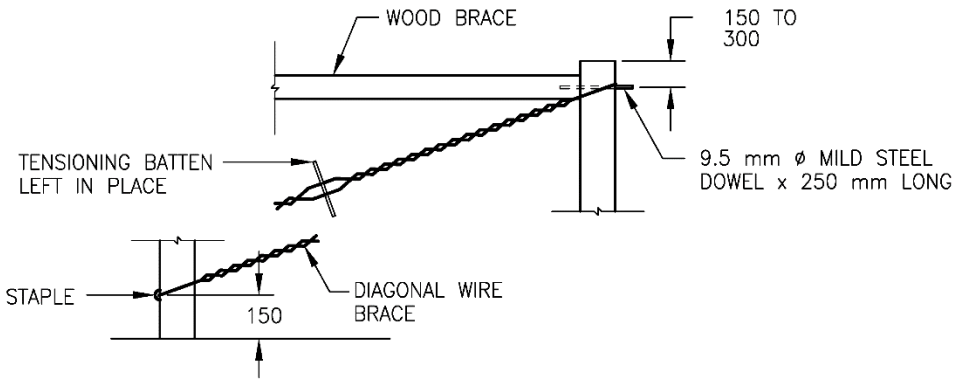
TYPE C, C1 & C2 FENCE STABILIZATION REQUIREMENTS | SP741–02.03



END BRACE
(BASIC END TENSIONING PANEL)



INTERMEDIATE BRACE
(BASIC INTERMEDIATE TENSIONING PANEL)



WOOD AND WIRE BRACE DETAIL

GENERAL NOTES:

BARBED WIRE SPOOL LENGTH 400 m.

STRAINING PANELS AND/OR BRACE PANELS ARE REQUIRED FOR TENSIONING WIRE AND AT SIGNIFICANT CHANGES IN FENCE ALIGNMENT.

POST SIZES: ALL POSTS – 120 mm (MIN.) DIAMETER EXCEPT:
LINE POSTS – 100 mm (MIN.) DIAMETER
CORNER POSTS – 140 mm (MIN.) DIAMETER

POST LENGTH: (EXCEPT LINE POSTS AND GATE POSTS) ALL POSTS TO BE THE SPECIFIED LINE POST LENGTH (SEE SP741–01.01) PLUS 400 mm FOR THE STABILIZING INCREASED DEPTH OF BURY.

BURY DEPTH: DEPTH OF END AND PANEL POSTS NORMALLY NOT LESS THAN 1200 mm. WHERE 1200 mm NOT ATTAINABLE BUT EXCEEDING 900 mm, USE DOUBLE TENSIONING STRAINING PANEL ASSEMBLIES AT NOT EXCEEDING 200 m SPACING. GREATER MINIMUMS ARE REQUIRED FOR PEAT SOILS, ETC.

POST SPACING AND WOOD BRACES:

WOOD BRACE	SIZE	LENGTH	POST SPACING APPROXIMATE
HORIZONTAL	80 mm	2.4 m	2.5 m
HORIZONTAL	100 mm	3.0 m	3.1 m
* DIAGONAL	100 mm	3.0 m	2.9 m

* NOTE: SOFT, MARSH AND/OR PEAT SOIL CONDITIONS MAY WARRANT DIAGONAL WOOD BRACES IN PLACE OF HORIZONTALS IN ADDITION TO THE LONGER POSTS.

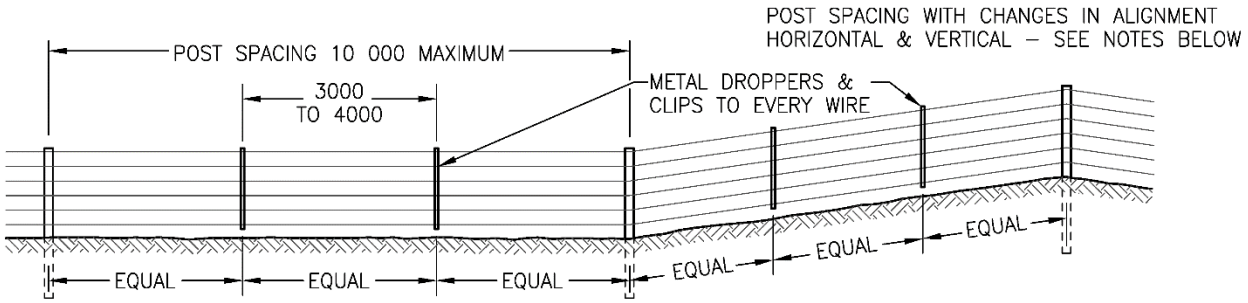
NOTE: ROCK, SHALE OR FRIABLE ROCK CONDITIONS SEE SP741–04.01 FOR METAL POSTS AND BRACES.

WIRE BRACES: 2 STRANDS OF 3.5 mm NOMINAL DIAMETER GALVANIZED WIRE WITH LOOPED ENDS AROUND DOWEL AND STAPLED TO POSTS – TWISTED TAUT ONE WAY OR BOTH WAYS AS INDICATED ABOVE.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

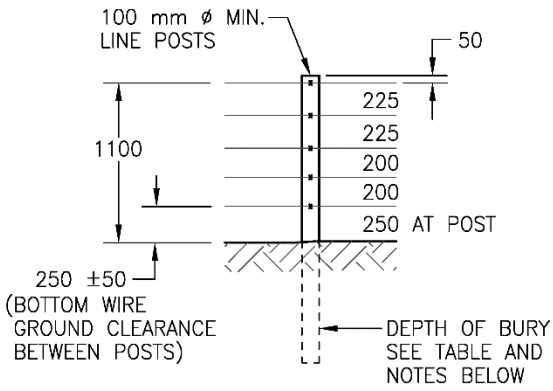
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HIGH–TENSILE FENCES (SMOOTH WIRE & WOOD POSTS) | SP741–03.01



HIGH–TENSILE SMOOTH–WIRE FENCING

FOR RANGE LAND AND OTHER LOCATIONS WITH FAVOURABLE TERRAIN, SOIL CONDITIONS AND FENCE ALIGNMENT.

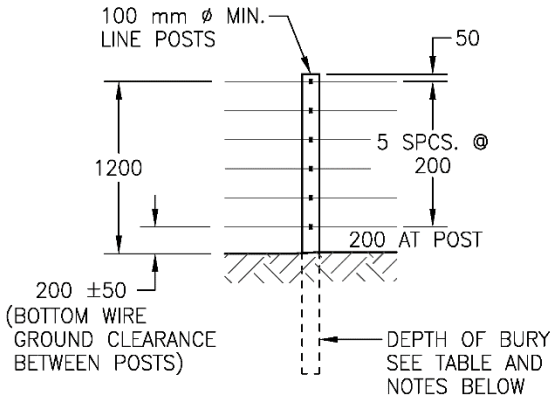


POST SPACING: 10 m MAX.
(HIGH SNOW AREAS 7 m MAX.)

	DEPTH OF BURY	POST LENGTH
FIRM SOIL	650 mm	1.8 m
MEDIUM CLAY	850 mm	2.0 m
PEAT SOIL	1050 mm	2.2 m

TYPE HT FENCE

RANGELAND AND WILD LIFE SENSITIVE AREAS



POST SPACING: 7 m MAX.

	DEPTH OF BURY	POST LENGTH
FIRM SOIL	750 mm	2.0 m
MEDIUM CLAY	950 mm	2.2 m
PEAT SOIL	1150 mm	2.4 m

TYPE HT–1 FENCE

HEAVY LIVESTOCK PRESSURES/FREEWAYS (WHERE PERMITTED OR REQUIRED)

NOTE: REDUCE LINE POST SPACING FOR SUCH AS NURSERY LIVESTOCK ENCLOSURES (HT–1 FENCE) AND AT ALL MINOR HORIZONTAL AND VERTICAL ALIGNMENT CHANGES. SEE SP741–03.03 FOR "SHALLOW", "MEDIUM" CURVES AND CORNER CHANGES IN ALIGNMENT.

GENERAL NOTES:

SETTING OUT AND CONSTRUCTION SHALL BE IN GENERAL ACCORDANCE WITH SECTION 741 AND MINISTRY OF AGRICULTURE AND FOOD PUBLICATION #ISBN 0–7719–9824–4 (BCMAF) PROCEDURES OR AS OTHERWISE REQUIRED.

POST AND BRACES: PRESSURE TREATED WOOD AS SPECIFIED IN SECTION 909.

WIRE: 2.5 mm NOMINAL DIAMETER HIGH–TENSILE GALVANIZED WIRE AS SPECIFIED BY SUBSECTION 316.07.

METAL DROPPERS AND CLIPS, SPLICERS AND FASTENERS & TENSIONING DEVICES. } PROPRIETRY MANUFACTURE TO SUBSECTION 316.12.

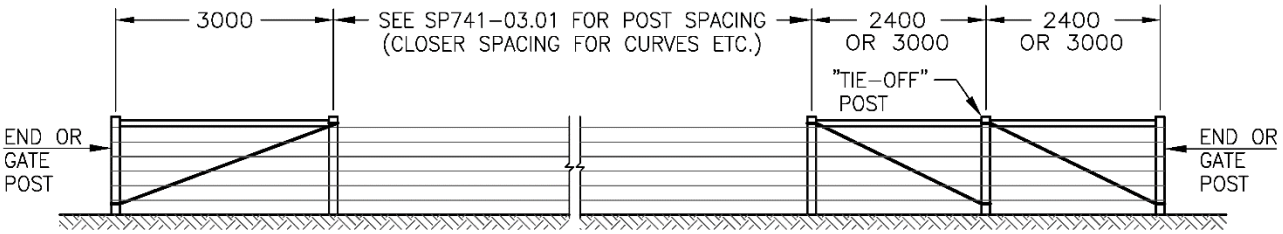
STAPLES: 45 mm LONG OF 3.5 mm NOMINAL DIAMETER GALVANIZED WIRE: DOUBLE STAPLE ON CURVES AND RISE AND DIP POSTS TO BCMAF RECOMMENDATIONS.

LINE POSTS: 100 mm MINIMUM DIAMETER SET FIRM BY DRIVING AND SET PLUMB EXCEPT FOR REQUIRED LEAN ON "MINOR" CURVES, SEE POST SIZE AND LENGTH FOR OTHER CURVES ON SP741–03.02 & SP741–03.03.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

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HIGH–TENSILE FENCE STABILIZATION | SP741–03.02

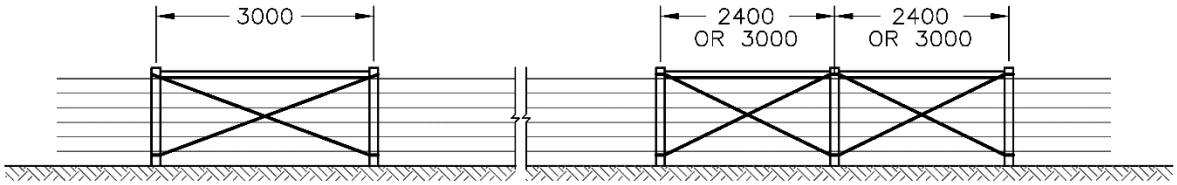


TERMINAL SINGLE ASSEMBLIES **TERMINAL DOUBLE ASSEMBLIES**

(FOR SCANT BURY DEPTH – SEE TABLE BELOW)

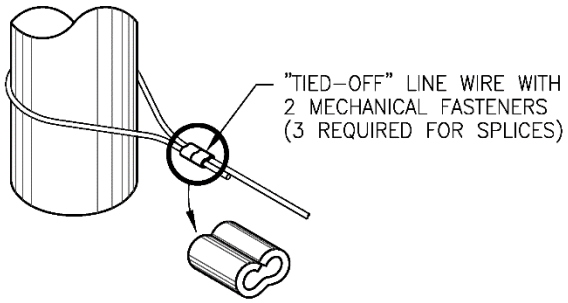
TERMINAL (END AND GATE) TENSIONING ASSEMBLIES

(CORNER AND INTERSECTION ASSEMBLIES SIMILAR)



INTERMEDIATE SINGLE ASSEMBLIES **INTERMEDIATE DOUBLE ASSEMBLIES**

INTERMEDIATE TENSIONING ASSEMBLIES



GENERAL NOTES:

COIL OF SMOOTH HIGH–TENSILE WIRE IN EXCESS OF 1000 m (APPROX. 100#)

POST SIZES: ALL 120 mm ø MINIMUM EXCEPT
LINE POSTS 100 mm ø MIN., CORNER POSTS 140 mm ø MINIMUM AND GATE POSTS 190 mm ø MINIMUM.
HORIZONTAL BRACES: 100 mm ø MIN., 3.0 m LONG (2.4 m LONG PERMITTED WITH DOUBLE ASSEMBLIES).

TERMINAL (END AND GATE) TENSIONING ASSEMBLIES SHALL BE PROVIDED TO TENSION EVERY LINE WIRE. WHERE TERMINAL SPACING IS IN EXCESS OF 1000 m *, INTERSPERSE WITH INTERMEDIATE TENSIONING ASSEMBLIES, UTILIZING ANY REQUIRED INTERSECTION AND CORNER ASSEMBLIES AT NOT EXCEEDING 1000 m * SPACINGS.
* SEE REDUCED SPACING REQUIREMENTS BELOW: NORMALLY 400 TO 600 m BUT MINIMUM 150 m.
FOR CONSTRUCTION OF SINGLE AND DOUBLE ASSEMBLIES SEE SP741–01.02, SP741–01.03, SP741–02.02 & SP741–02.03 BUT USE HIGH TENSILE WIRE FOR WIRE BRACES.

DEPTH OF BURY AND POST LENGTH FOR SINGLE AND DOUBLE ASSEMBLIES.

	HT (5 WIRE) FENCE				HT–1 (6 WIRE) FENCE			
	SINGLE		DOUBLE		SINGLE		DOUBLE	
TYPICAL SOIL TYPE	BURY DEPTH (mm)	LENGTH (m)	BURY DEPTH (mm)	LENGTH (m)	BURY DEPTH (mm)	LENGTH (m)	BURY DEPTH (mm)	LENGTH (m)
FIRM SOIL	1050	2.2	850	2.0	1150	2.4	950	2.2
MEDIUM CLAY	1250	2.4	1050	2.2	1350	2.6	1150	2.4
PEAT SOIL	–	–	1250	2.4	–	–	1350	2.6

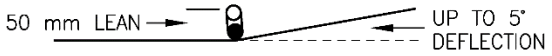
REDUCE SPACING OF TENSIONING ASSEMBLIES WHERE ABOVE BURY DEPTHS ARE NOT ADEQUATE FOR THE PARTICULAR SOIL CONDITION AND FOR CHANGES IN HORIZONTAL ALIGNMENT AT CURVES AND UNTIED–OFF CORNERS:
3 m REDUCTION FOR EVERY 1° OF DEFLECTION CHANGE
NOTE: ALL "CURVE" POSTS TO BE 120 mm ø MIN. WITH LENGTHS AND BURY AS FOR END POSTS.

REDUCTION OF ASSEMBLY SPACING FOR UNEVEN TERRAIN CHANGE IN VERTICAL ALIGNMENT:
– 50 m PER HUMP AND DIP AND 100 m PER MAJOR HUMP AND DIP.

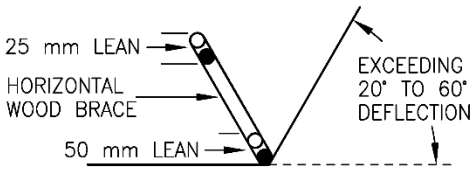
NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

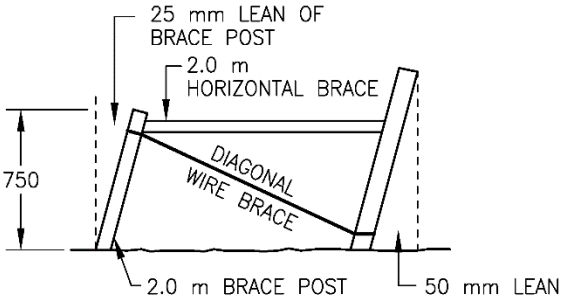
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HIGH–TENSILE FENCE STABILIZATION | SP741–03.03

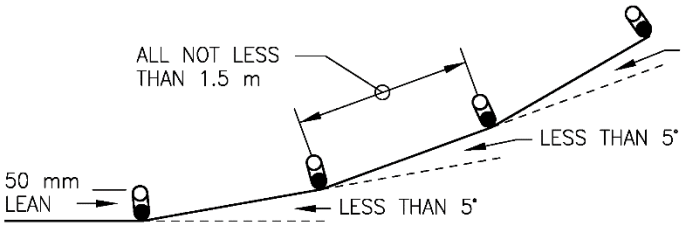


MINOR CURVE

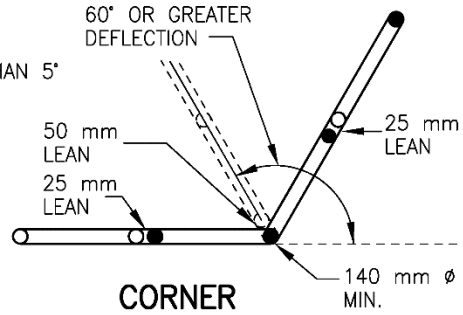




MEDIUM CURVE



SHALLOW CURVE



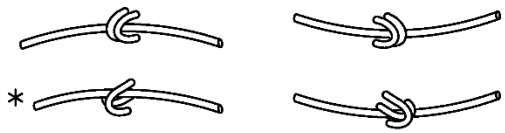
CORNER

NOTE: BRACES ON OUTSIDE OF CURVE WHERE NECESSARY WITH OPPOSITE LEAN AND WIRE BRACE.

TENSIONING DEVICES IN EVERY MECHANICALLY FASTENED "TIED–OFF" LINE WIRE. AT MID–SPAN OR TOWARDS END WITH THE MAJORITY OF ALIGNMENT CHANGES, BETWEEN ADJOINING ASSEMBLIES (FOR STRAIGHT AND LEVEL FENCE RUNS UNDER 200 m: DEVICES NEXT TO ONE TENSIONING ASSEMBLY).

DIP POSTS TO BE PROVIDED WITH STEEL POST(S) OR FOOTING BLOCKS TO BCMAF RECOMMENDATIONS TO RESIST POST LIFTING IN HOLLOWS.

DOUBLE STAPLE ALL LINE WIRES AT CURVES, CORNERS, DIPS AND RISES AS INDICATED BELOW AND AS INDICATED * FOR HEAVY SNOW AREAS.

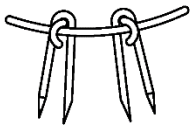


RISE POST

DIP POST



SHALLOW CORNER

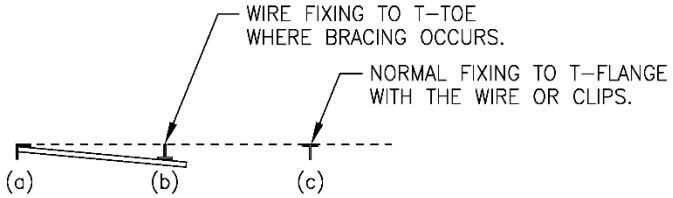


RIGHT ANGLE

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

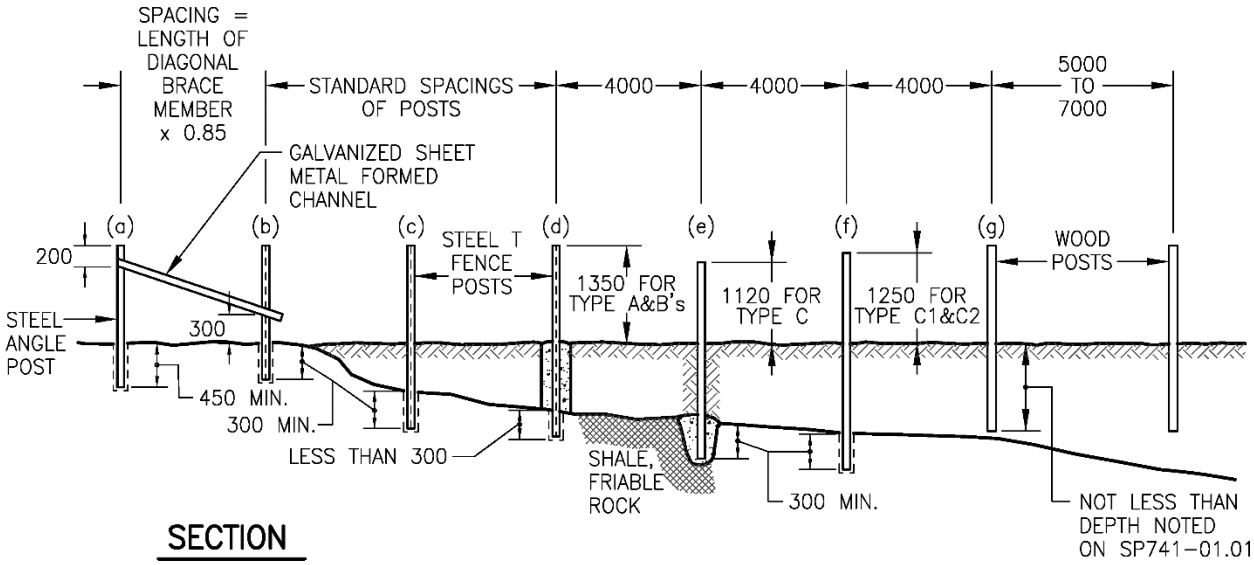
BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

TYPE A, B, B1, C, C1 & C2 FENCE ON OR OVER ROCK | SP741–04.01



PLAN

NOTE:
FOR MINOR ROCK OUTCROPPING AND UNFORESEEN MINIMUM OVERBURDEN OCCURRENCES WIRE TIED MULTIPLE WOODPOST/BRACE ASSEMBLIES MAY BE APPROVED BY THE MINISTRY REPRESENTATIVE.



SECTION
EXPOSED ROCK AND OVERBURDEN CONDITIONS

NOTES:

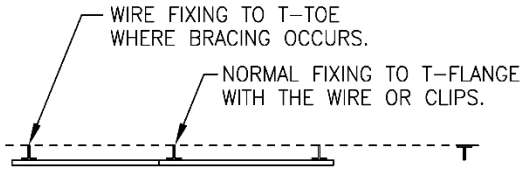
STEEL ANGLE END POST (a) AND SIMILAR CORNER AND INTERSECTION POSTS SET IN CEMENT AND SAND NON–SHRINK MORTAR WITH STEEL BRACE(S) TO ADJOINING STEEL FENCE POST(S) (b) SECURELY AND BOLTED EACH END.

STEEL TEE FENCE POSTS SHALL BE USED ON:
- ROCK (b)
- MINIMUM OVERBURDEN (c) OR (d)
- WITH OVERBURDEN (f) LESS THAN THE NORMAL BURY
- DEPTH FOR WOOD POSTS REQUIRED ON SP741–01.01
- TO SHALE, LOOSE OR FRIABLE ROCK (e)
- ENLARGED DRILLED HOLE SET IN CEMENT MORTAR.

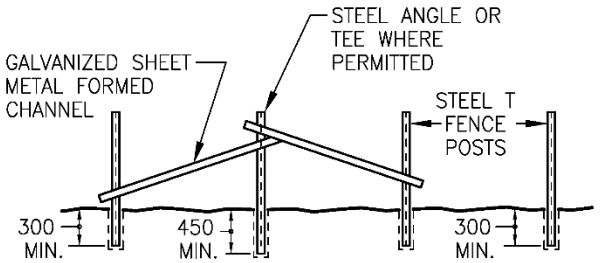
WHERE DEPTH OF BURY OF A STEEL POST INTO SOLID ROCK IS LESS THAN (a) & (b) ABOVE AS AT (d) CONCRETE FOOTING (4:2:1 MIX) OF ADEQUATE SIZE MAY BE APPROVED BY THE MINISTRY REPRESENTATIVE.

CURVED ALIGNMENT SPACING OF STEEL POSTS SEE SP741–04.02.

STEEL ANGLE AND TEE POSTS CHANNEL BRACES ARE SPECIFIED BY SUBSECTION 316.09 AND INSTALLATION BY SUBSECTION 741.34.



PLAN



SECTION
INTERMEDIATE TENSIONING ASSEMBLY

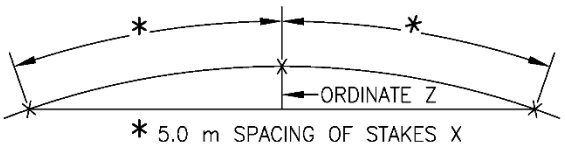
NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

TYPE A, B, B1, C, C1 & C2 FENCE SPECIAL CONDITIONS | SP741–04.02

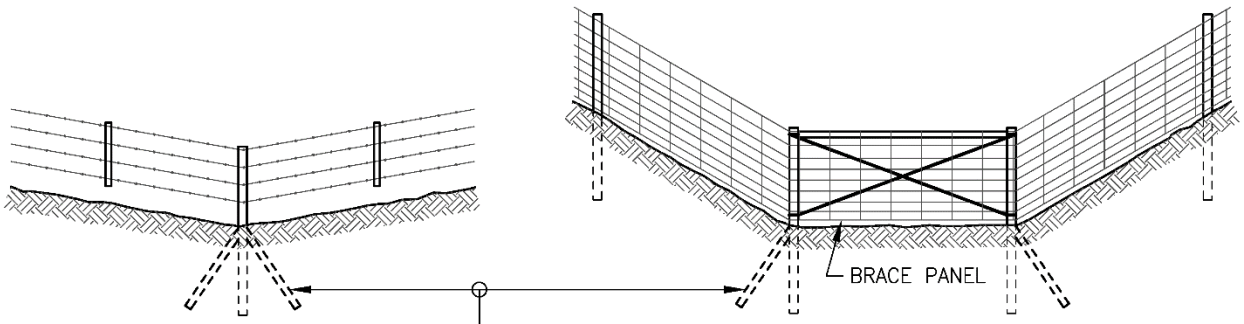
FENCE POST SPACING (m)

ORDINATE Z (mm)	WOOD POST	STEEL POST
LESS THAN 100	5.0	4.0
GREATER THAN 100 TO 150	4.5	3.6
GREATER THAN 150 TO 200	4.0	3.2
GREATER THAN 200 TO 350	3.2	2.6
GREATER THAN 350	2.4	2.0



CURVED HORIZONTAL ALIGNMENTS

REDUCED LINE POST SPACING



GRADUAL ABRUPT

NOTES

1. BRACE PANELS OR STRAINING PANELS, (SEE SP741–01.03 & SP741–02.03) ARE REQUIRED ON EITHER SIDE OF AN ABRUPT DEPRESSION TO RELIEVE FENCING TENSION AND MINIMIZE UPLIFT TENDENCY.
2. KEEP POST OR ASSEMBLY OUT OF WET CONDITIONS.

CHANGE OF VERTICAL ALIGNMENT

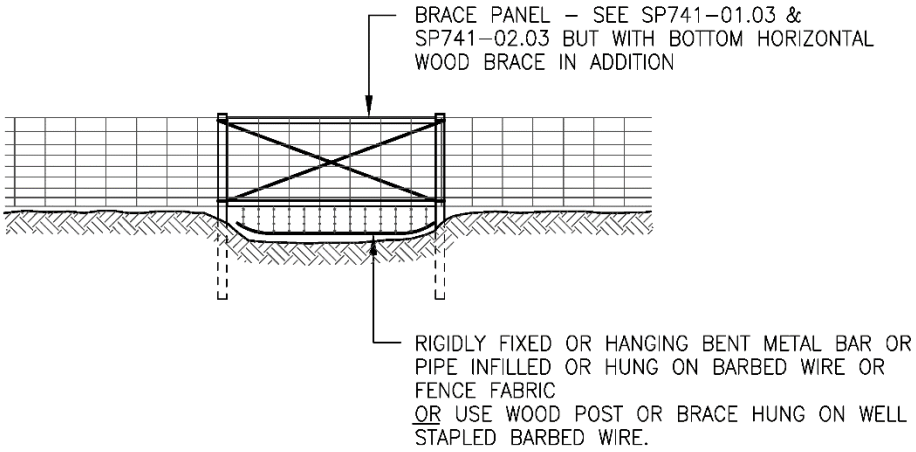
DEPRESSION – AS SHOWN

SUMMIT – SIMILAR EXCEPT NO HOLD–DOWN ANCHORS REQUIRED.

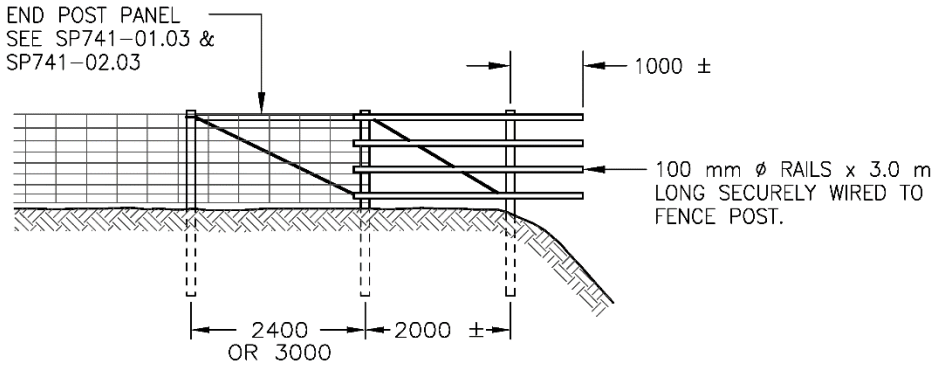
NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

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TYPE A, B, B1, C, C1 & C2 FENCE SPECIAL CONDITIONS	SP741–04.03



CONTINUOUS FENCING
(AT DITCHES, ETC.)



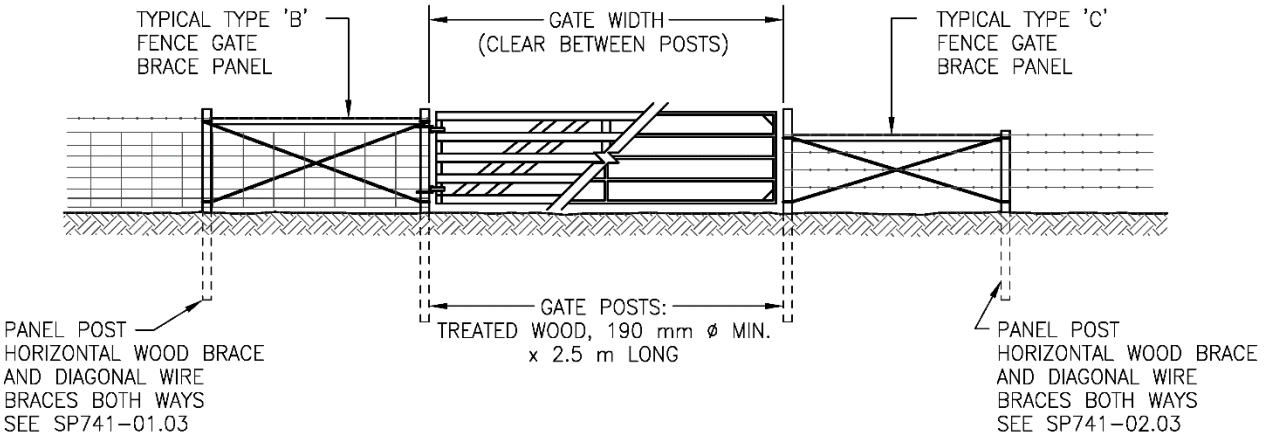
INTERRUPTED FENCES
(AT GULLEY OR CREEK)

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

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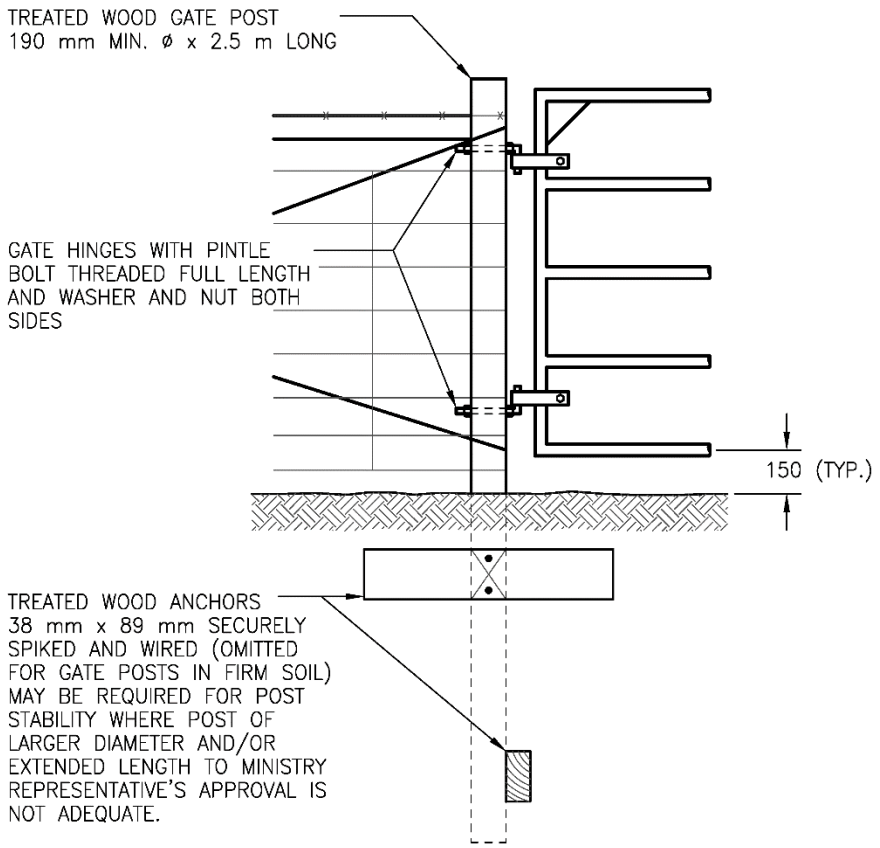
FARM AND RANGE GATES

SP741–04.04



GATES IN TYPE A, B, B1, C, C1 & C2 FENCES

(SEE SP741–05.03 FOR TYPE D FENCE GATE)



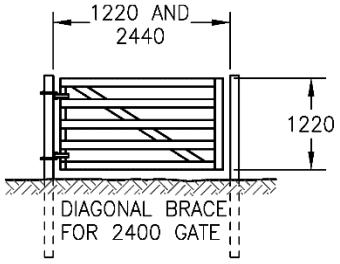
TYPICAL HINGE DETAILS

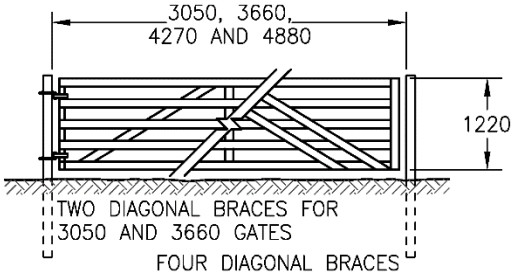
NOTE: SEE SP741–04.05 FOR GATE TYPES AND APPLICATIONS

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

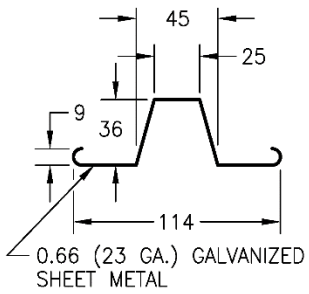
FARM AND RANGE GATES

SP741–04.05

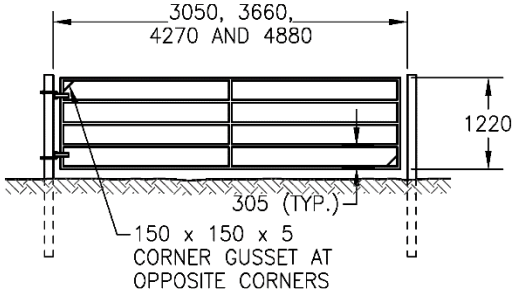




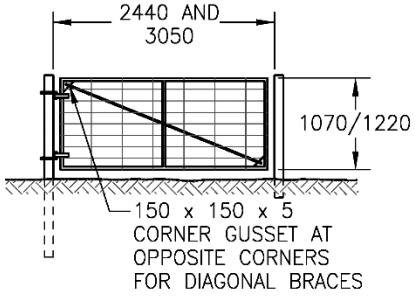
TYPE 1 GATE
(SHEET METAL MEMBERS)

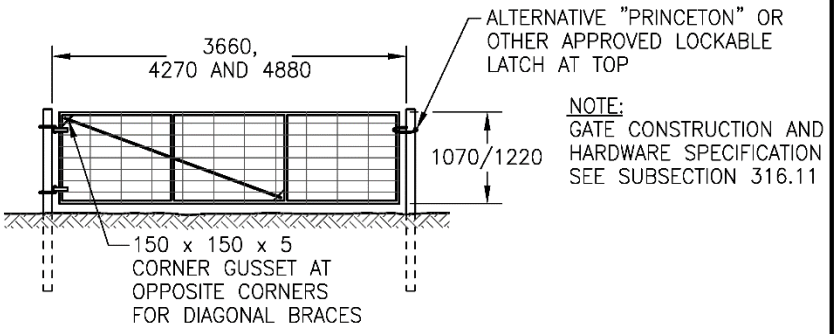


GATE MEMBER

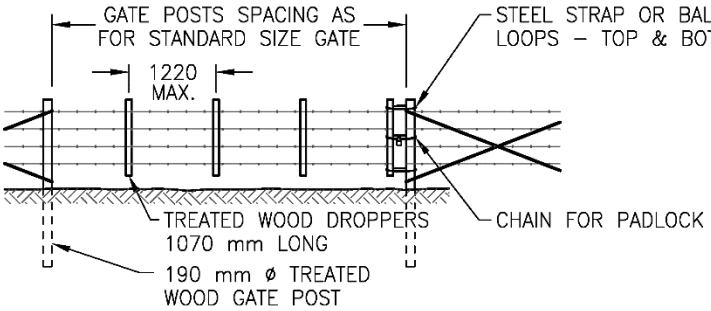


TYPE 2 GATE
(ALL PIPE)





TYPE 3 GATE
(PIPE AND WIRE FABRIC INFILL)



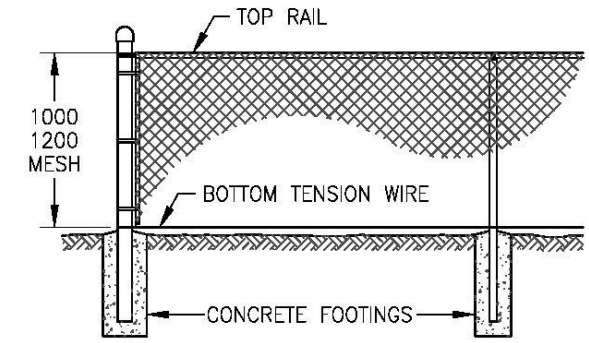
TYPE 5 GATE
(RANGE SLIP–WIRE)

NOTE:
SEE SP741–04.04 FOR GATE DETAILS

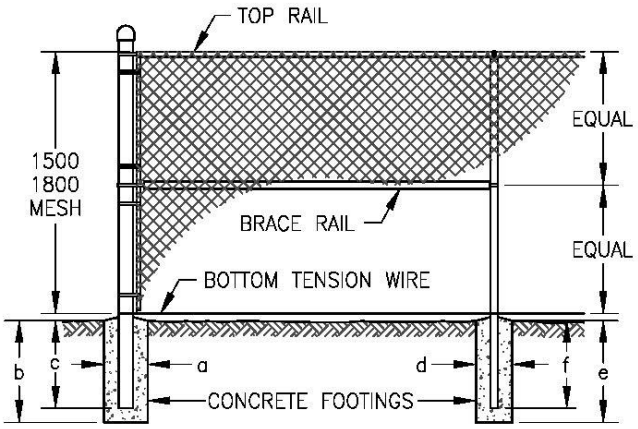
NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

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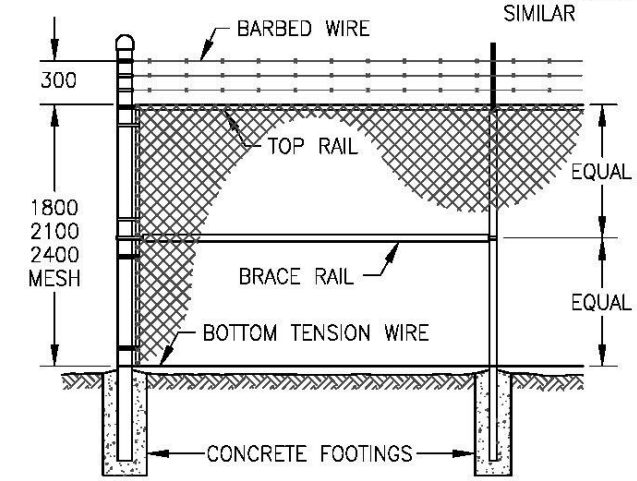
TYPE D – CHAIN LINK FENCE – GENERAL LAYOUT | SP741–05.01



RESIDENTIAL HEIGHT
(1.0 OR 1.2 m)



STANDARD HEIGHT
(1.5 OR 1.8 m)



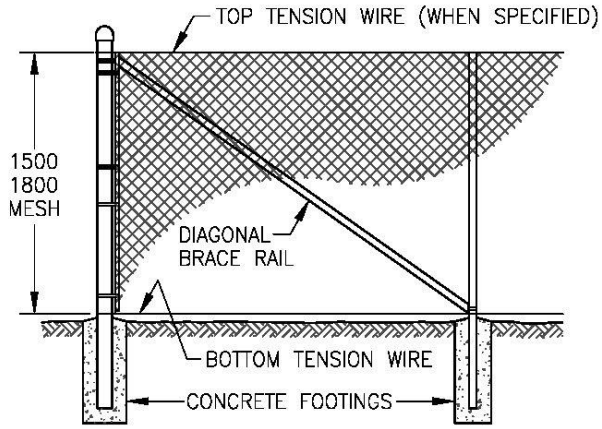
SECURITY HEIGHT
(1.8, 2.1 OR 2.4 m
EXCLUSIVE OF BARBED WIRE HEIGHT)

CONCRETE FOOTINGS

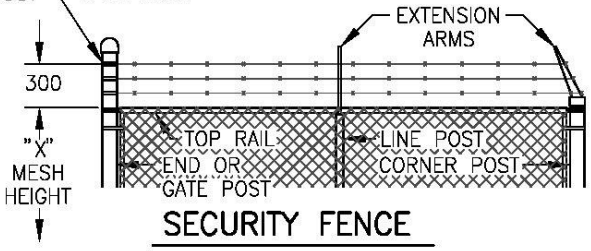
CONCRETE TO CONFORM TO SS 218, CLASS C OR 20 MPa GROUT.

MESH HEIGHT "X" (mm)	CONCRETE FOOTINGS SIZE * (mm) TERMINAL POSTS a	b	c	LINE POSTS d	e	f
1000	200	750	600	200	600	450
1200	200	750	600	200	600	450
1500	300	1050	900	250	750	600
1800	300	1050	900	250	750	600
2100	350	1050	900	300	900	750
2400	350	1050	900	300	900	750

* INCREASE FOR PEAT SOIL ETC. FOR ROCK AND OTHER CONDITIONS SEE SUBSECTION 741.36.01(iii)



NOTE:
INTERMEDIATE TENSION ASSEMBLIES WITH TOP TENSION WIRE REQUIRE DIAGONAL BRACE ON EACH SIDE.



SECURITY FENCE
(EXTENSION ARMS AND EXTRA HEIGHT END, GATE AND STRAINING POSTS FOR 3 STRANDS OF BARBED WIRE)

GENERAL NOTES:

MATERIAL SPECIFICATIONS FOR PIPE, MESH AND BARBED WIRE SEE SECTION 316.
MESH FABRIC SELVAGE EDGE:
- TWISTED EDGE UP FOR 1.8 m OR GREATER HEIGHT EXCEPT
- KNUCKLE EDGE UP FOR 1.8 m HEIGHTS FOR SCHOOL, PARK OR PLAYGROUNDS
- KNUCKLE SELVAGE BOTH EDGES FOR 1.5 m AND LESS.

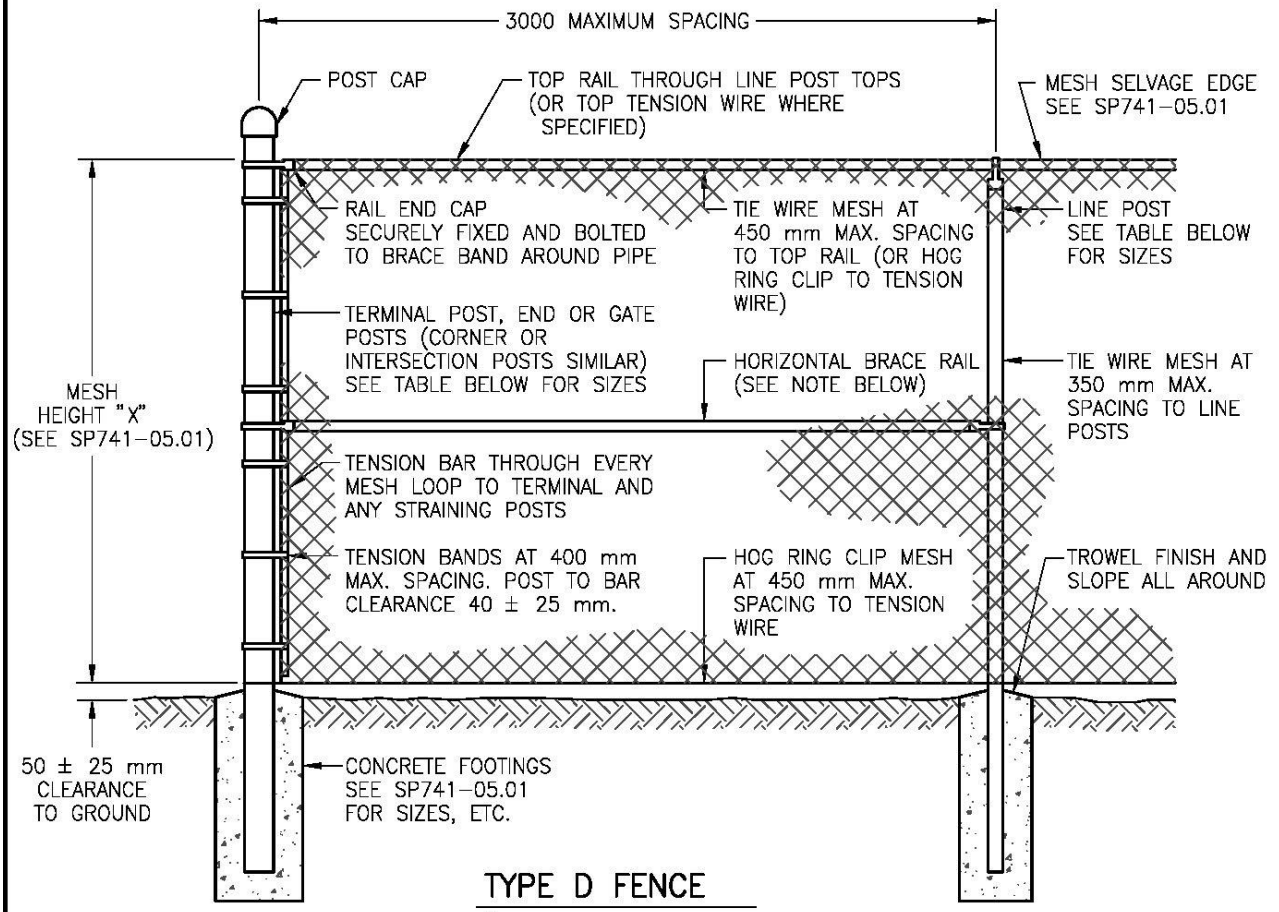
ACCESSORY MATERIAL SEE SP741–05.02.
INSTALLATION, INCLUDING STRAINING POST REQUIREMENTS SEE SUBSECTION 741.36.
GATE DETAILS SEE SP741–05.03 & SP741–05.04.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

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TYPE D – CHAIN LINK FENCE COMPONENTS

SP741–05.02



TYPE D FENCE

(CHAIN LINK WIRE MESH AND STEEL PIPE RAILS. RESIDENTIAL, STANDARD AND SECURITY HEIGHTS SEE SP741–05.01)

ACCESSARY MATERIAL

POST AND RAIL CAPS: OF APPROVED DESIGN OF PRESSED STEEL OR CAST OR MALLEABLE IRON AND GALVANIZED MINIMUM 600 g/m^2, OR CAST ALUMINUM.

GALVANIZING COATING: TO STEEL WIRE, CLIPS, BARS AND BANDS, MINIMUM 360 g/m^2.

TENSION WIRE: BOTTOM AND TOP WHEN SPECIFIED NOMINAL 5.0 mm ø (6 ga.) MEDIUM TENSILE GALVANIZED WIRE: NOMINAL 3.75 mm ø (9 ga.) FOR RESIDENTIAL HEIGHT FENCES.

TIE WIRE: NOMINAL 3.75 mm (9 ga.) ALUMINUM WIRE FOR MESH FIXING TO TOP RAIL AND LINE POSTS.

HOG RING CLIPS: NOMINAL 3.75 mm (9 ga.) GALVANIZED STEEL WIRE CLIPS FOR MESH FIX TO BOTTOM (OR TOP) TENSION WIRE.

TENSION BAR: MINIMUM 5 mm x 16 mm GALVANIZED MILD STEEL FLAT BAR.

TENSION OR BRACE BANDS: MINIMUM 3 mm x 16 mm GALVANIZED FORMED MILD STEEL FLATBARS WITH GALVANIZED BOLTS AND NUTS FOR ALL RAIL END CAP AND TENSION BAR FIXING.

POST AND RAIL SIZES – O.D. MINIMUM (mm)

	RESIDENTIAL	STANDARD	SECURITY
LINE POSTS	38	48	60
END, CORNER & STRAINING POSTS	48	60	73
TOP RAILS	33	33	42
HORIZONTAL BRACE RAILS*	––	33	42

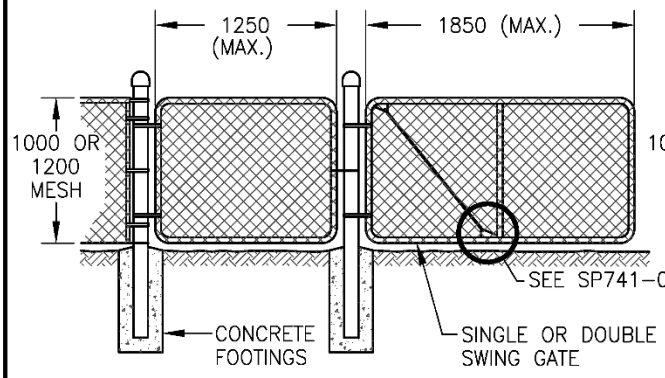
*NOTE: DIAGONAL PIPE BRACES MAY BE SPECIFIED WHERE SOIL CONDITIONS WARRANT ESPECIALLY WHERE TOP TENSION WIRE IS CALLED FOR IN PLACE OF TOP RAIL. FOR GATE POST SIZES SEE SP741–05.04.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

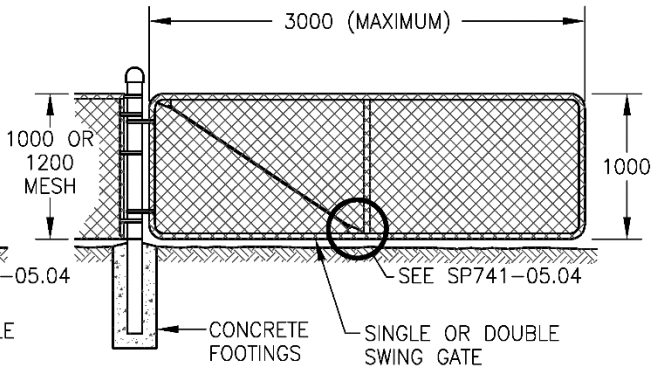
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TYPE D – CHAIN LINK FENCE GATES

SP741–05.03



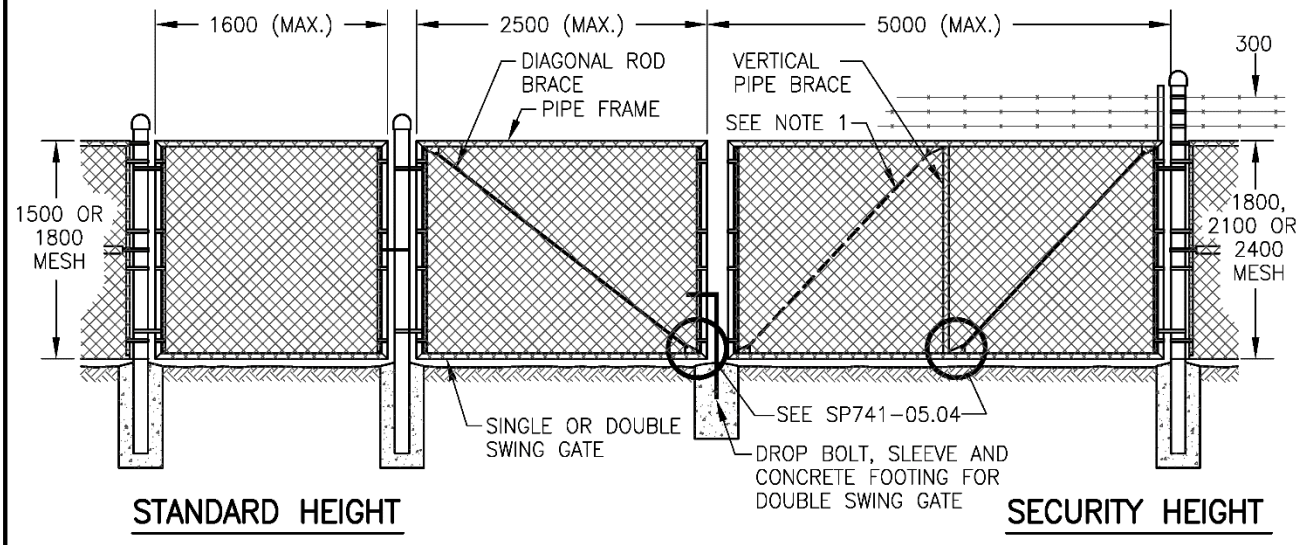
GATES NOT EXCEEDING 1.85 m WIDTH OF OPENING
- GATE FRAME: 33 mm O.D. GALVANIZED TUBING 1.9 kg/m
- VERTICAL PIPE BRACE: 27 mm O.D. GALVANIZED TUBING 1.5 kg/m
- DIAGONAL BRACE: 5.0 mm NOMINAL DIAMETER GALVANIZED TENSION WIRE TO 5 mm CLEATS AS ON SP741–05.04.



GATES NOT EXCEEDING 3.0 m WIDTH OF OPENING
- GATE FRAME: 33 mm O.D. GALVANIZED TUBING 2.5 kg/m
- VERTICAL BRACE: 27 mm O.D. GALVANIZED PIPE 1.7 kg/m
- DIAGONAL BRACE: 9.5 mm NOMINAL DIAMETER GALVANIZED ROD WITH HOOKED ENDS TO 5 mm CLEATS AS ON SP741–05.04

RESIDENTIAL HEIGHT GATES

(TYPE IV – SEE SUBSECTION 316.11.05)



STANDARD HEIGHT

SECURITY HEIGHT

- GATE FRAME: 42 mm OD GALVANIZED PIPE 3.4 kg/m.
- VERTICAL PIPE BRACE: 33 mm OD GALVANIZED PIPE 2.5 kg/m
- DIAGONAL BRACE: 9.5 mm DIAMTER GALVANIZED ROD TO 5 mm CLEATS AS SHOWN ON SP741–05.04.

NOTE 1: SECOND DIAGONAL ROD BRACE REQUIRED FOR ALL GATES OVER 3.0 m TO 5.0 m WIDTH OF OPENINGS: REVERSE SLOPE OF SECOND ROD FOR ALL SLIDING GATES.

STANDARD AND SECURITY HEIGHT GATES

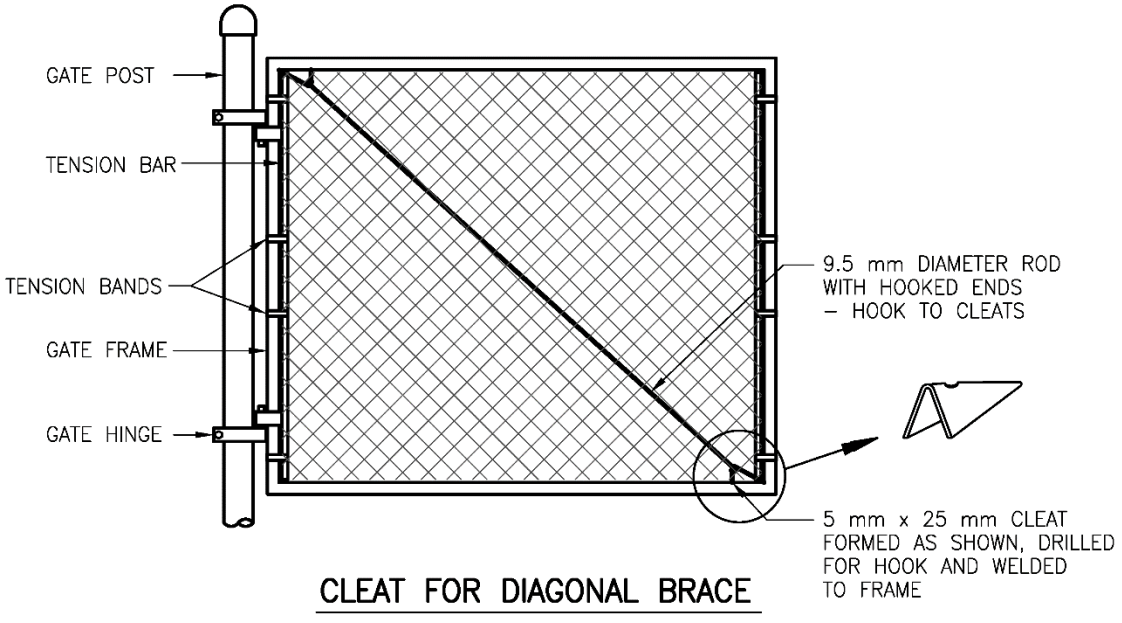
(TYPE IV – SEE SUBSECTION 316.11.05)

NOTE:
FOR FENCE GATE DETAILS SEE SP741–05.04.

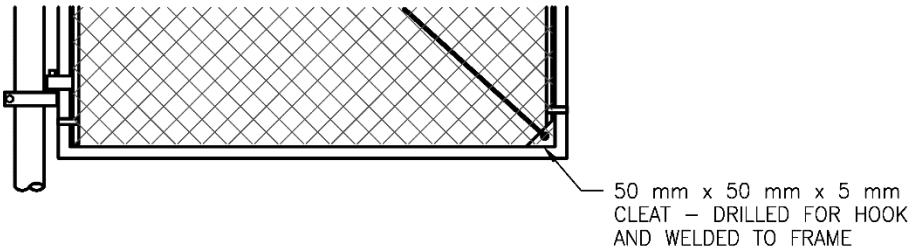
NOT TO SCALE
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TYPE D – CHAIN LINK FENCE GATE DETAILS | SP741–05.04



CLEAT FOR DIAGONAL BRACE



ALTERNATE CLEAT FOR DIAGONAL BRACE

GATE POST SIZES – NOMINAL OUTSIDE DIAMETER

RESIDENTIAL		48 mm
STANDARD HEIGHT FENCE GATE	NOT EXCEEDING 2.5 m	60 mm
	EXCEEDING 2.5 m	73 mm
SECURITY HEIGHT FENCE GATE	NOT EXCEEDING 2.5 m	73 mm
	EXCEEDING 2.5 m	89 mm

CONCRETE FOOTING SIZES – UP TO 2.5 m WIDE GATES SEE SP741–05.01.
– OVER 2.5 m WIDTH, FOOTINGS 400 mm ø, 1250 mm DEEP

SECURITY HEIGHT FENCE GATES TYPE 4

SIMILAR TO STANDARD HEIGHT BUT WITH EXTENDED VERTICAL STILES (AND BRACE) FOR BARBED WIRE OVER.

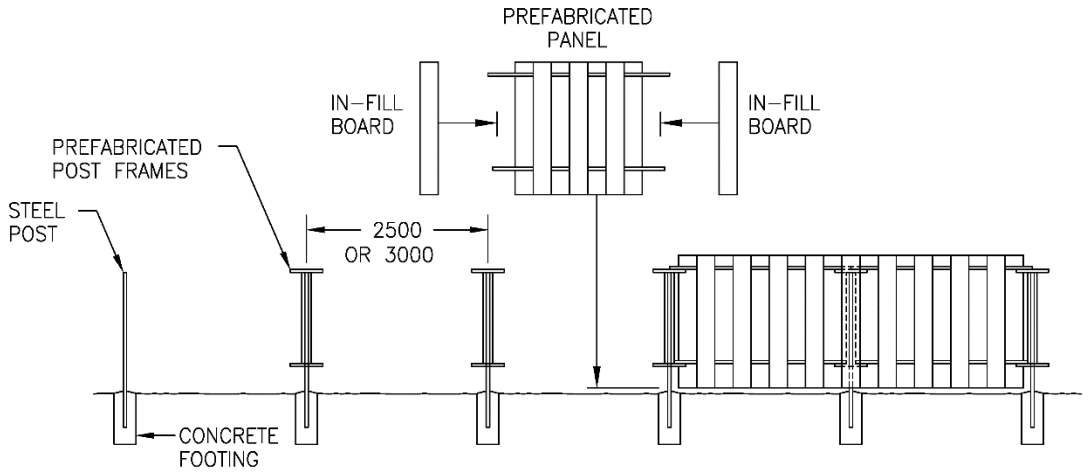
GENERAL NOTES

- MATERIAL SPECIFICATIONS FOR PIPE, MESH AND BARBED WIRE, SEE SECTION 316.
- CAPS AND ACCESSORY MATERIAL, SEE SP741–05.02.
- HARDWARE, SEE SUBSECTION 316.11.05.
- SLIDING GATE MAY BE DETAILED FOR GATE OPENINGS OVER 4.0 m WITH ROLLING HARDWARE AS REFERRED TO IN SUBSECTION 316.11.05.

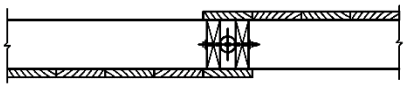
NOT TO SCALE
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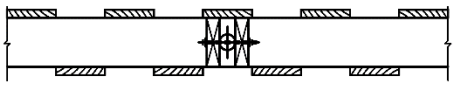
WOOD FENCE – PRIVACY – GENERAL ARRANGEMENT | SP741–06.01



GENERAL ARRANGEMENT
ELEVATION

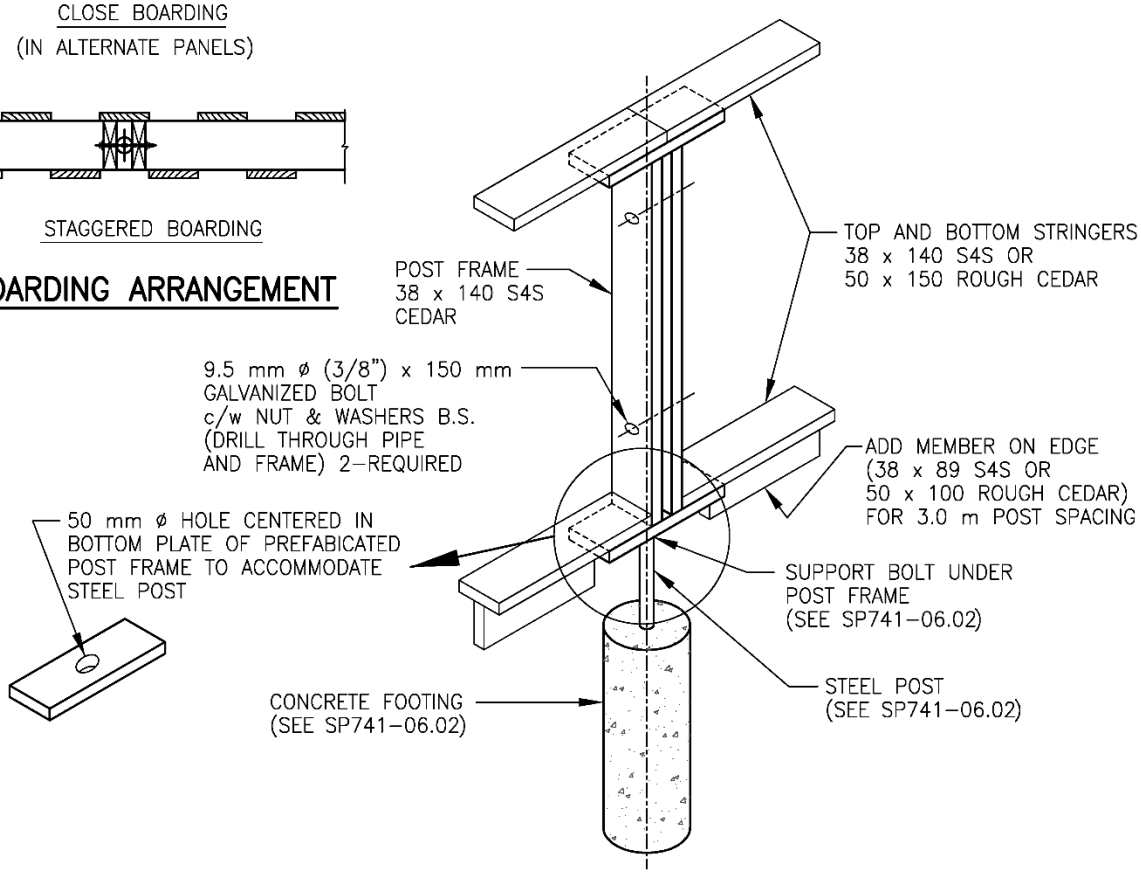


CLOSE BOARDING
(IN ALTERNATE PANELS)



STAGGERED BOARDING

BOARDING ARRANGEMENT



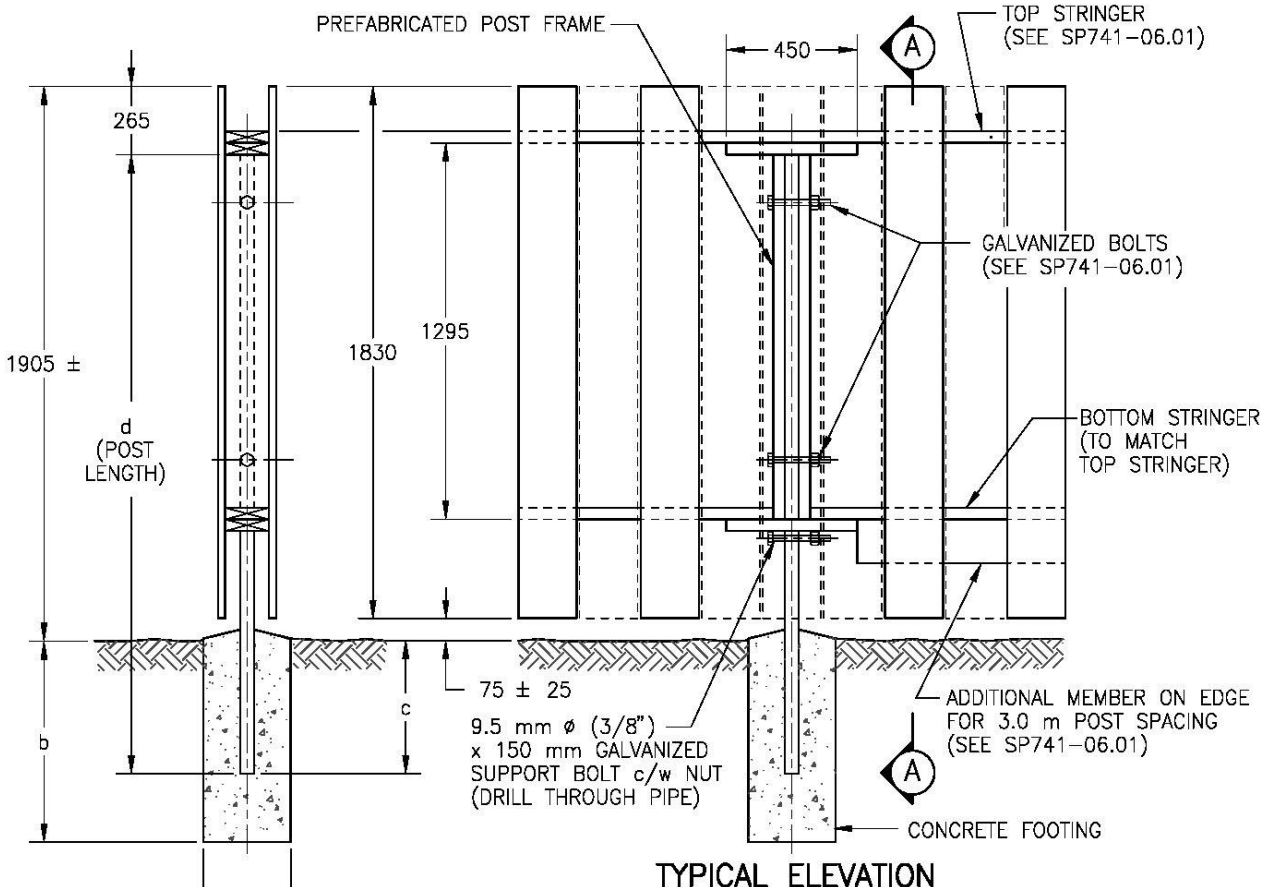
ISOMETRIC DETAIL

NOT TO SCALE
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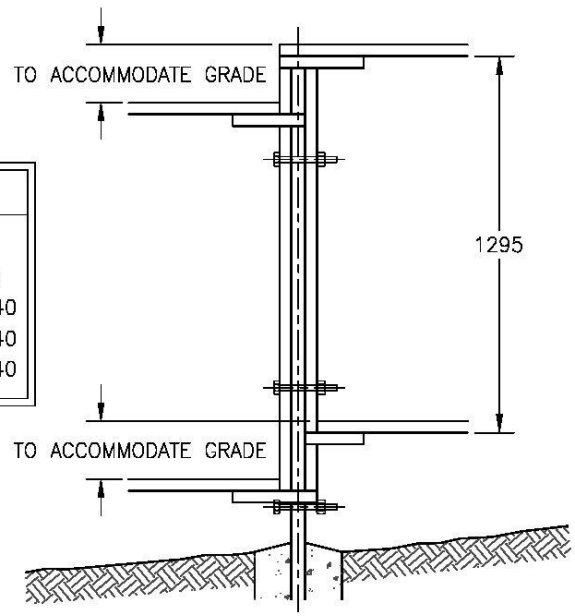
WOOD FENCE – PRIVACY – DETAILS

SP741–06.02



TYPICAL ELEVATION

SECTION A–A



DETAIL – STEPPED POST FRAME

CONCRETE FOOTING AND POST

POST SPACING	2.5 m				3.0 m			
	a	b	c	d	a	b	c	d
FIRM SOIL	200	750	500	2140	250	750	500	2140
STANDARD	250	750	500	2140	300	750	500	2140
PEAT SOIL	300	750	500	2140	300	1000	800	2440

CONCRETE FOOTINGS:

CONCRETE TO CONFORM TO SS 218, CLASS C OR 20 MPa GROUT.

STEEL POSTS:

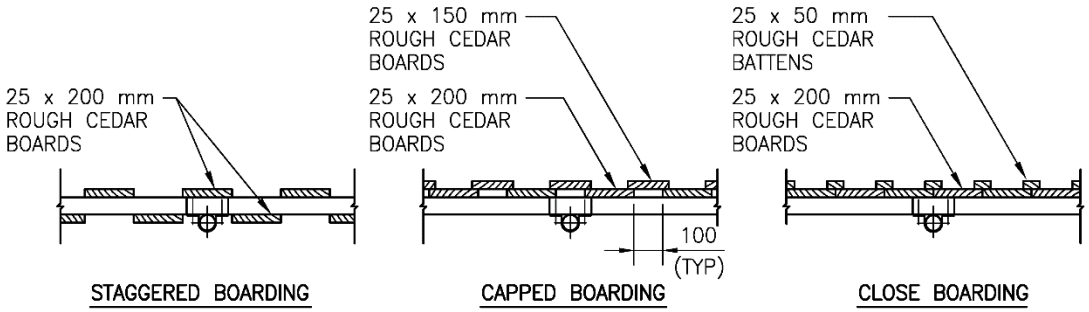
POSTS TO BE 48 mm O.D. GALVANIZED STANDARD WEIGHT TUBING.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

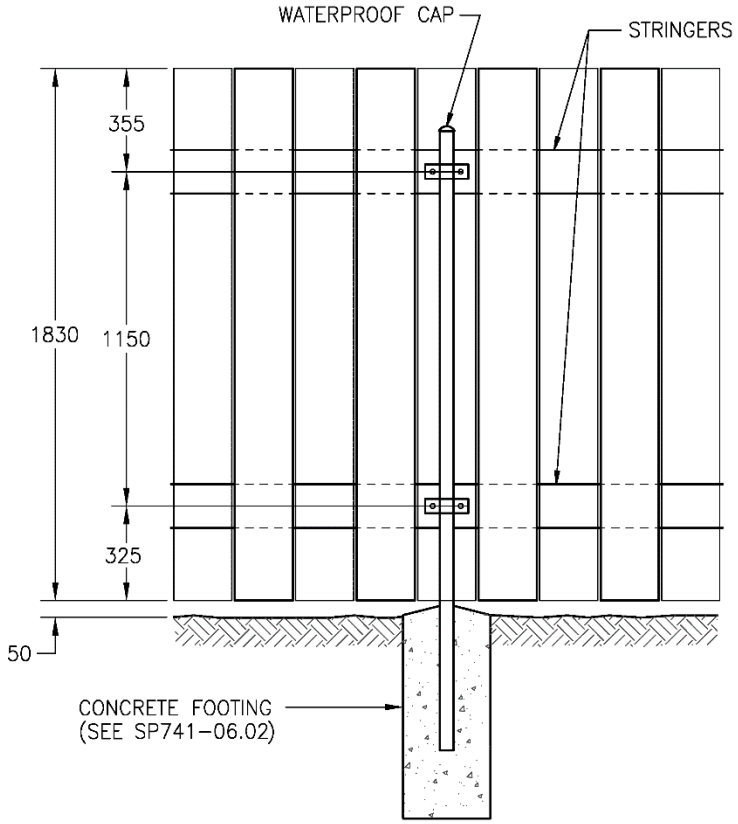
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WOOD FENCE – PRIVACY – GENERAL ARRANGEMENT

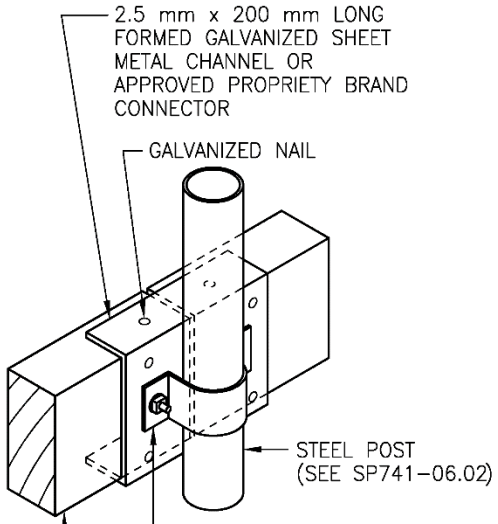
SP741–06.03



BOARDING ARRANGEMENT



TYPICAL ELEVATION



STRINGERS: (TYP.)

50 x 100 ROUGH CEDAR FOR 2.5 m POST SPACING
50 x 150 ROUGH CEDAR FOR 3.0 m POST SPACING

ISOMETRIC DETAIL

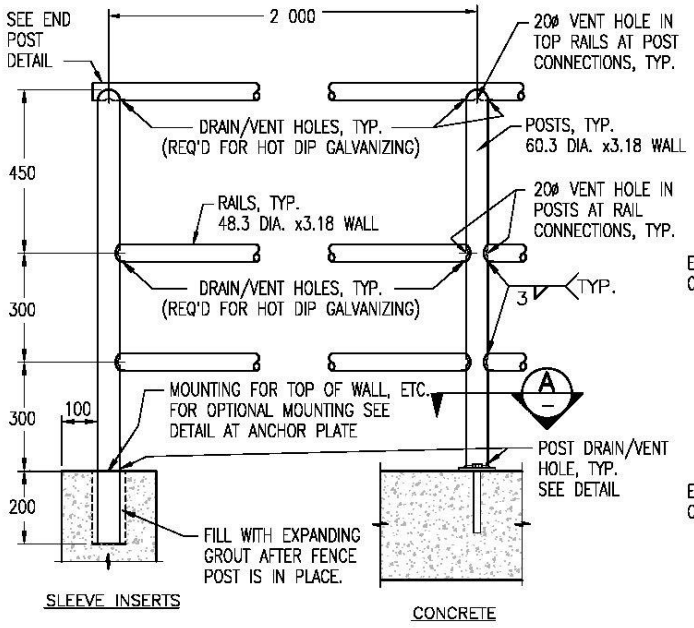
NOT TO SCALE
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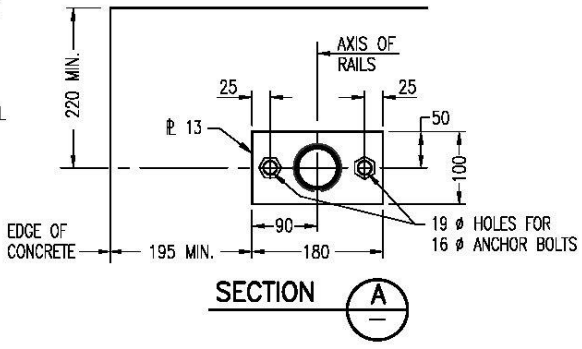
PEDESTRIAN SIDEWALK FENCE – WELDED

SUITABLE FOR LOCATIONS WHERE GATHERING OF MANY PEOPLE IMPROBABLE

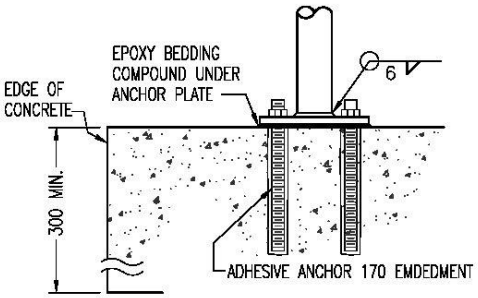
SP741–07.01



WELDED FENCE



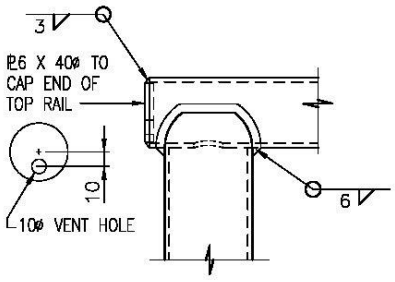
SECTION A



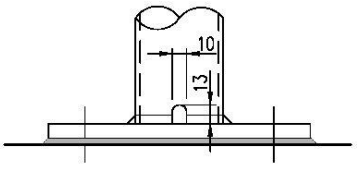
DETAIL AT ANCHOR PLATE

NOTES:

1. ALL RAIL AND POSTS TO BE CAN/CSA G40.21 GRADE 350W OR ASTM A500 GRADE C. MISCELLANEOUS PLATE TO BE CAN/CSA G40.21 GRADE 300W.
2. ALL ROUGH EDGES SHALL BE GROUND SMOOTH AND WELD SPLATTER SHALL BE REMOVED.
3. WELDED FENCE SHALL BE GALVANIZED AFTER FABRICATION IN ACCORDANCE WITH ASTM A123/A123M.
4. FIELD WELDING SHALL BE KEPT TO A MINIMUM. DAMAGED GALVANIZING SHALL BE REPAIRED IN ACCORDANCE WITH SS 741.39.
5. VENT AND DRAIN HOLES:
   A. RAILS SHALL BE PROVIDED WITH A BOTTOM DRAIN/VENT HOLE AT BOTH ENDS BETWEEN POSTS.
   B. WHERE A POST IS EMBEDDED IN CONCRETE WITH A SLOPED TOP SURFACE OR WHERE IT IS SUPPORTED ON A SLOPED BASE PLATE, THE DRAIN/VENT HOLE NEAR THE BOTTOM OF THE POST SHALL BE LOCATED ON THE LOWER SIDE OF THE SLOPE.
6. POST ANCHORS SHALL BE ASTM A307 THREADED ROD, GALVANIZED TO THE REQUIREMENTS OF ASTM A153/A153M.
7. ACCEPTABLE ADHESIVES FOR BONDING THE ANCHORS INTO CONCRETE INCLUDE:
   A. HILITI HIT HY–200
   B. REDHEAD EPCON C6+
   C. UCAN FLO–ROX FR6–SD
   D. POWERS PE1000+
8. ANCHORS SHALL BE INSTALLED IN ACCORDANCE WITH THE ADHESIVE MANUFACTURE'S INSTRUCTIONS.
9. HOLES IN CONCRETE FOR ANCHORS MUST BE DRILLED WITH A HAMMER DRILL AND A CARBIDE TIPPED BIT. CONCRETE IN HOLES MUST BE WATER–SATURERATED OR DRY WHEN INSTALLING ANCHORS.
10. ALTERNATIVE ANCHOR INSTALLATIONS THAT DO NOT MEET THE DETAILS ON THIS DRAWING WILL REQUIRE AN ENGINEERED DESIGN DEMONSTRATING THAT THE ALTERNATIVE INSTALLATION DETAILS ARE CAPABLE OF CARRYING THE REQUIRED LOADS. NATIONAL BUILDING CODE OF CANADA LOADING REQUIREMENTS ON "GUARDS" AND ON "HANDRAILS" SHALL BE CONSIDERED IN DETERMINING POST ANCHOR LOADING.



END POST DETAIL



DRAIN/VENT HOLE DETAIL

TYPICAL AT ENDS OF POSTS AND RAILS, HOLES FOR POSTS TO BE ON THE SAME AXIS AS RAILS

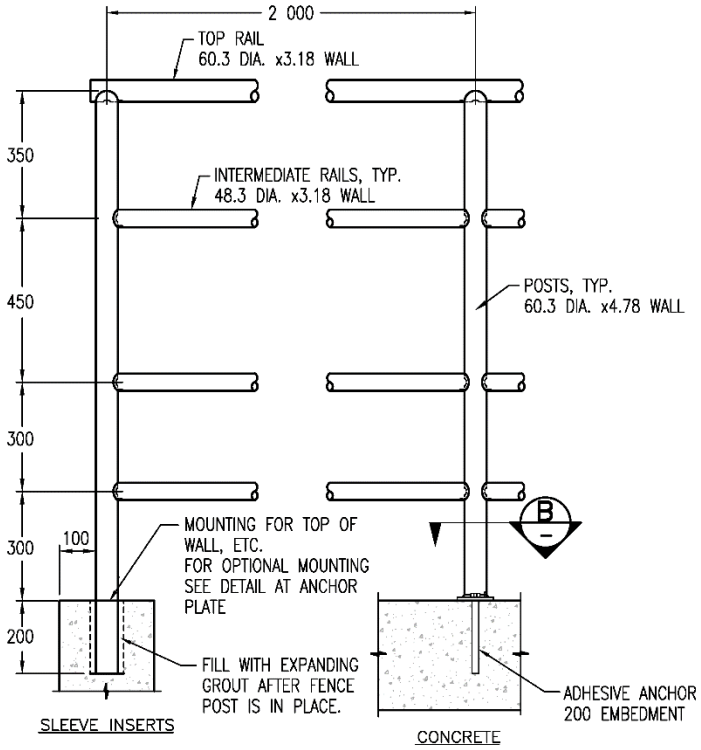
NOT TO SCALE
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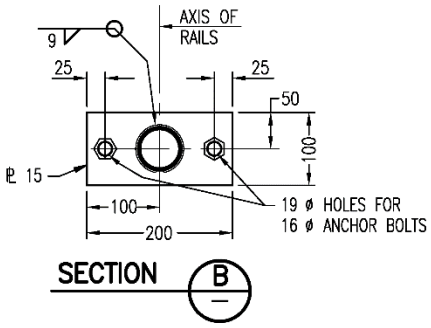
BICYCLIST SIDEWALK FENCE – WELDED

SUITABLE FOR LOCATIONS WHERE GATHERING OF MANY PEOPLE IMPROBABLE

SP741–07.02



WELDED FENCE



SECTION B

NOTES:

1. ALL DETAILS NOT SHOWN SIMILAR TO SP741–07.01.
2. SEE NOTES ON SP741–07.01.

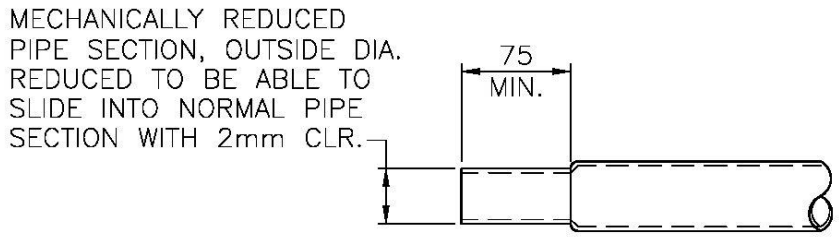
NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

No.	Revision	Date
A	GENERAL REVISIONS, SLIP JOINT FENCE DELETED	FEB 2016

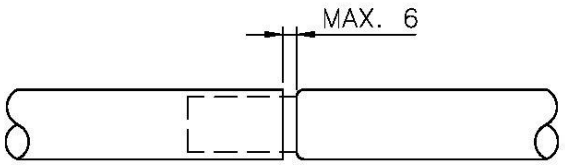
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FENCE – WELDED OR SLIP–ON RAIL SPLICE

SP741–07.03



SWAGE PIPE END



INSTALLATION OF SWAGE PIPE END

NOTE: SPLICE SPACING TO BE APPROX. 7 TO 9m, WITH RAIL TO BE CONTINUOUS OVER A MINIMUM OF TWO POSTS.

NOT TO SCALE
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BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

SECTION 751

TOPSOIL AND LANDSCAPE GRADING

DESCRIPTION

751.01 Scope – This Section refers to those portions of work that are unique to the supply and placement of topsoil and subsequent finish grading. In this Section, the term "topsoil" is used to identify:

(a) On-Site topsoil: material stockpiled for use, or

(b) Imported topsoil, or

(c) Manufactured topsoil (Growing Medium), including Biosolids Growing Medium (BGM) or Biotic Soil Matrix (BSM) products where specified in the Contract.

751.02 Related Work

- SS 754, Planting of Trees, Shrubs and Ground Covers;
- SS 757, Revegetation Seeding.

751.03 References

- Canadian Landscape Standard,
- Canadian System of Soil Classification,
- Canada Fertilizers Act and Fertilizers Regulations,
- Canadian National Master Specification,
- Weed Control Act & Weed Control Regulation,
- British Columbia. Field Sampling Manual,
- British Columbia Environmental Laboratory Manual
- Environmental Management Act & Organic Matter Recycling Regulation, and
- Open Burning Smoke Control Regulation.

751.04 Topsoil Supplied by the Contractor – The Contractor shall advise the Ministry Representative of the sources of topsoil not less than seven days before any is used in the work.

The Contractor shall, at the Contractor's expense, acquire a soil analysis from an accredited soil testing laboratory, to verify that supplied material is within the requirements indicated. All soil tests shall follow the British Columbia Field Sampling and British Columbia Environmental Laboratory Manual. Results of the soil test are to be submitted to the Ministry Representative prior to installation.

The Ministry Representative will approve all topsoil once it has met the standard required at the source.

MATERIALS

751.11 General – In this Section, a range of measurable physical and chemical properties are set out as being acceptable in a topsoil. Compliance with this Section shall be determined by testing for those properties. When imported or on-Site topsoil is used, it shall be tested and modified as necessary by a mixture of other components to bring its properties to within the range set in SS 751.16, or as stated in the Special Provisions. Topsoil shall not be prepared or handled in an excessively wet or frozen condition, or in any manner in which structure is adversely affected.

751.12 Topsoil Types – Three topsoil types are described in SS 751.13 through SS 751.15. Regardless of origin, all types shall conform to SS 751.16.

751.13 On-Site Topsoil – On-Site topsoil may be used, as specified in the Special Provisions, provided that it meets the standard set for imported topsoil and can be modified to meet the requirements set out for the specified topsoil. On-Site topsoil shall be defined as the existing "A" horizon containing accumulated organic matter. On-Site topsoil shall be tested prior to stockpiling. Upon approval by the Ministry Representative of the suitability of the on-Site topsoil for topsoil, a sufficient quantity of stripped on-Site topsoil shall be stockpiled where shown on the Drawings or in areas designated for stockpiling.

751.14 Imported Topsoil – Imported topsoil shall be of a sandy loam or loamy sand texture (no less than 50% sand by weight) containing between 4% and 15% organic matter (dry weight basis).

Imported topsoil shall be free of propagules of plant species designated as noxious under the Weed Control Act & Regulation, and other invasive or undesirable plant species, as determined by the Ministry Representative.

At least 80% of imported topsoil shall pass a Tyler #10 sieve after appropriate crushing of structural units using accepted laboratory test methods.

751.15 Manufactured Topsoil – Manufactured topsoil is any soil or growing medium mixture with chemical and physical properties that fall within ranges required by this Section for a particular application.

Manufactured topsoil shall meet the requirements in the Organic Matter Recycling Regulation and conform to SS 751.16 or as specified in the Special Provisions.

751.15.01 Biotic Soil Media (BSM) – Where specified in the Contract, the BSM shall be designed as an alternative to topsoil and/or compost to accelerate development of depleted soils/substrates with low organic matter, low nutrient levels and limited biological activity.

BSM products used must be non-toxic with fibers that have been phytosanitized to eliminate potential weed seeds and pathogens prior to the introduction of soil building components.

BSM products shall not be used for erosion control and shall be applied to enhance vegetation establishment and growth only.

Where erosion control is required, the BSM shall be installed in a two-phase process with the BSM hydraulically applied first with the seed, then covered with an erosion control product. BSM products may also be applied dry with broadcast seeding methods. Refer to SS 757.

751.16 Requirements for Topsoil – Acceptance of commercially processing and mixing of topsoil components shall only be from a facility operating under a valid permit under the Environmental Management Act. The resulting product shall be a homogenous mixture having the required properties throughout.

The general amendment of any topsoil by mixing in situ with rototill cultivation equipment or using equipment with tracks to ensure mixing and good contact with subsurface soils after placement, is acceptable if carried out to the satisfaction of the Ministry Representative. Fertilizers typically used at time of seeding or planting will be surface applied and incorporated as described in SS 751.33.

The Contractor shall also require the laboratory to include recommendations for incorporating fertilizers and other soil amendments into the topsoil as needed for plant establishment and maintenance, and as they specifically relate to:

- grassed areas;
- ground covers, shrubs and trees;
- container or planter box installations; and,
- to site conditions and season of planting.

Refer to Table 751-A for required properties of growing medium for different applications.

751.16.01 Specific Requirements – Other specific requirements are as follows:

(a) Fertility:

(i) Nitrogen – total nitrogen shall be 0.2% to 0.6% by weight.

(ii) Phosphorus – available phosphorus shall be 20 to 250 ppm.

(iii) Potassium – available potassium shall be 50 to 1000 ppm.

(iv) Boron – concentration in saturation extract shall not exceed 1 ppm

(b) General:

(i) Acidity – in accordance with Table 751-A. Maximum of 0.5 kg/m^2 of dolomite lime to achieve the required pH level.

(ii) Salinity – saturation extract conductivity shall not exceed 3.0 mmho/cm at 25°C. Sodium absorption ration (SAR) as calculated from analysis of saturated extract shall not exceed 8.0.

(iii) C/N Ratio – carbon to nitrogen ratio shall not exceed 40:1.

(iv) Texture – in accordance with Table 751-A.

(v) Organic Content – in accordance with Table 751-A.

(vi) Sawdust – Cedar or redwood sawdust shall not be present in the topsoil.

Table 751-A: Properties of Topsoil for Different Applications

Properties	Low Traffic Lawn Areas Trees and Large Shrubs	Planting Areas, Planters, Shrub and Groundcover Areas
Texture: by particle size classes[1]	Percent of Dry Weight Mineral Fraction (%)	
• Coarse Gravel > 19 mm < 40 mm	0 – 3	0 - 3
• All Gravel > 2 mm < 40 mm	0 – 10	0 - 10
• Sand (0.05 to 2 mm)	30 – 70	30 – 70
• Silt & Clay combined	Maximum 60%	Maximum 60%
Acidity (pH)	6.0 – 7.0	4.5 – 7.0
Drainage: Minimum saturated hydraulic conductivity (cm/hr) in place	2.0	2.0
Organic Content: Percent of Dry Weight (%)	2 – 10	5 – 20

[1] Note: Per the Canadian Landscape Standard and Canadian System of Soil Classification